Sre-final-project

First created the containing folder for my project in Ubuntu.

```
Joro_21@L-Mejia21: ~/project
Joro_21@L-Mejia21:~$ ls
Documents  Projects  kubectl  minikube-linux-amd64  projects
Joro_21@L-Mejia21:~$ cd projects/
Joro_21@L-Mejia21:~/projects$ ls
sre-academy-training-main
Joro_21@L-Mejia21:~/projects$ mkdir sre-final-project
Joro_21@L-Mejia21:~/projects$ cd sre-final-project/
Joro_21@L-Mejia21:~/projects/sre-final-project$
```

Then moved to start creating the files required in order to mount everything and have it working:

```
Joro_21@L-Mejia21: ~/project
Joro_21@L-Mejia21:~/projects/sre-final-project$ ls
Joro_21@L-Mejia21:~/projects/sre-final-project$ mkdir app
Joro_21@L-Mejia21:~/projects/sre-final-project$ mkdir k8s
Joro_21@L-Mejia21:~/projects/sre-final-project$ mkdir monitoring
Joro_21@L-Mejia21:~/projects/sre-final-project$ ls
app  k8s  monitoring
Joro_21@L-Mejia21:~/projects/sre-final-project$
```

While choosing the App type, I thought: Why make it a Hello World style, when I can get me something that can have more alerts, like a simple calculator? More alerts because it has more Endpoints, and simple because it still maintains an easy to containerize model.

So by creating a new file within "app" directory, using the command: nano app/main.py, then entered generic code:

```
from fastapi import FastAPI, HTTPException

from fastapi.responses import Response

from prometheus_client import Counter, generate_latest


app = FastAPI(title="SRE Calculator API")


# Métricas Prometheus

calc_requests_total = Counter(

    "calc_requests_total", "Total calculator requests", ["operation"]

)


@app.get("/")

def home():

    return {"status": "ok", "message": "Calculator API running!"}


@app.get("/add")

def add(a: float, b: float):

    calc_requests_total.labels(operation="add").inc()

    return {"operation": "add", "a": a, "b": b, "result": a + b}


@app.get("/subtract")

def subtract(a: float, b: float):

    calc_requests_total.labels(operation="subtract").inc()
```

```
    return {"operation": "subtract", "a": a, "b": b, "result": a - b}


@app.get("/multiply")
def multiply(a: float, b: float):
    calc_requests_total.labels(operation="multiply").inc()
    return {"operation": "multiply", "a": a, "b": b, "result": a * b}


@app.get("/divide")
def divide(a: float, b: float):
    if b == 0:
        raise HTTPException(status_code=400, detail="Division by zero")
    calc_requests_total.labels(operation="divide").inc()
    return {"operation": "divide", "a": a, "b": b, "result": a / b}


@app.get("/metrics")
def metrics():
    return Response(generate_latest(), media_type="text/plain")
```

Joro_21@L-Mejia21: ~/project

```
Joro_21@L-Mejia21:~/projects/sre-final-project$ nano app/main.py
Joro_21@L-Mejia21:~/projects/sre-final-project$ ls -l
total 12
drwxr-xr-x 2 Joro_21 Joro_21 4096 Nov 20 18:29 app
drwxr-xr-x 2 Joro_21 Joro_21 4096 Nov 20 18:18 k8s
drwxr-xr-x 2 Joro_21 Joro_21 4096 Nov 20 18:18 monitoring
Joro_21@L-Mejia21:~/projects/sre-final-project$ cd app
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ ls
main.py
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ cat main.py
from fastapi import FastAPI, HTTPException
from fastapi.responses import Response
from prometheus_client import Counter, generate_latest

app = FastAPI(title="SRE Calculator API")

# Métricas Prometheus
calc_requests_total = Counter(
    "calc_requests_total", "Total calculator requests", ["operation"]
)

@app.get("/")
def home():
    return {"status": "ok", "message": "Calculator API running!"}

@app.get("/add")
def add(a: float, b: float):
    calc_requests_total.labels(operation="add").inc()
    return {"operation": "add", "a": a, "b": b, "result": a + b}

@app.get("/subtract")
def subtract(a: float, b: float):
    calc_requests_total.labels(operation="subtract").inc()
    return {"operation": "subtract", "a": a, "b": b, "result": a - b}

@app.get("/multiply")
def multiply(a: float, b: float):
    calc_requests_total.labels(operation="multiply").inc()
    return {"operation": "multiply", "a": a, "b": b, "result": a * b}

@app.get("/divide")
```

```
@app.get("/divide")
def divide(a: float, b: float):
    if b == 0:
        raise HTTPException(status_code=400, detail="Division by zero")
    calc_requests_total.labels(operation="divide").inc()
    return {"operation": "divide", "a": a, "b": b, "result": a / b}

@app.get("/metrics")
def metrics():
    return Response(generate_latest(), media_type="text/plain")
Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

Then, created another dir with the requirements:

```
Joro_21@L-Mejia21: ~/project

Joro_21@L-Mejia21:~/projects/sre-final-project/app$ nano requirements.txt
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ cat requirements.txt
fastapi
uvicorn
prometheus_client

Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

And last but not least, the Dockerfile:

```
  GNU nano 6.2
FROM python:3.11-slim

WORKDIR /app
COPY . .

RUN pip install --no-cache-dir -r requirements.txt

EXPOSE 8000
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

```
Joro_21@L-Mejia21: ~/project
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ nano requirements.txt
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ cat requirements.txt
fastapi
uvicorn
prometheus_client

Joro_21@L-Mejia21:~/projects/sre-final-project/app$ nano Dockerfile
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ ls
Dockerfile  main.py  requirements.txt
Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

Moved to Visual Studio for better view and editing options (For yaml files and others).

First error encountered was a failure in docker integration with WSL, easy fix: Opened Docker Desktop, then reviewed settings in Docker Desktop, as recommended by ubuntu:

```
PROBLEMS 3   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS 2          bash - app
Joro_21@L-Mejia21:~/projects/sre-final-project$ cd app/
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ docker build -t alejoro21/sre-calculator:1.0 .

The command 'docker' could not be found in this WSL 2 distro.
We recommend to activate the WSL integration in Docker Desktop settings.

For details about using Docker Desktop with WSL 2, visit:

https://docs.docker.com/go/wsl2/

Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

However, error was that I had not logged into Docker:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ docker build -t alejoro21/sre-calculator:1.0 .

Joro_21@L-Mejia21:~/projects/sre-final-project/app$ docker login
Authenticating with existing credentials... [Username: alejoro21]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ docker build -t alejoro21/sre-calculator:1.0 .
[+] Building 28.0s (10/10) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                            0.1s
 => => transferring dockerfile: 216B                                                                            0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                             2.6s
 => [auth] library/python:pull token for registry-1.docker.io                                                   0.0s
 => [internal] load .dockerignore                                                                               0.1s
 => => transferring context: 2B                                                                                 0.0s
 => [1/4] FROM docker.io/library/python:3.11-slim@sha256:193fdd0bbcb3d2ae612bd6cc3548d2f7c78d65b549fcaa8af75624  8.0s
 => => resolve docker.io/library/python:3.11-slim@sha256:193fdd0bbcb3d2ae612bd6cc3548d2f7c78d65b549fcaa8af75624  0.1s
 => => sha256:1771569cc1299abc9cc762fc4419523e721b11a3927ef968ae63ba0a4a88f2da 251B / 251B                      0.3s
 => => sha256:b3dd773c329649f22e467ae63d1c612a039a0559dec99ffb9ada904ab5c60c55 14.36MB / 14.36MB                2.1s
 => => sha256:22b63e76fde1e200371ed9f3cee91161d192063bcff65c9ab6bf63819810a974 1.29MB / 1.29MB                  0.8s
 => => sha256:0e4bc2bd6656e6e004e3c749af70e5650bac2258243eb0949dea51cb8b7863db 29.78MB / 29.78MB                2.8s
 => => extracting sha256:0e4bc2bd6656e6e004e3c749af70e5650bac2258243eb0949dea51cb8b7863db                       2.9s
 => => extracting sha256:22b63e76fde1e200371ed9f3cee91161d192063bcff65c9ab6bf63819810a974                       0.3s
 => => extracting sha256:b3dd773c329649f22e467ae63d1c612a039a0559dec99ffb9ada904ab5c60c55                       1.6s
 => => extracting sha256:1771569cc1299abc9cc762fc4419523e721b11a3927ef968ae63ba0a4a88f2da                       0.1s
 => [internal] load build context                                                                               0.1s
 => => transferring context: 1.64kB                                                                             0.0s
 => [2/4] WORKDIR /app                                                                                          0.4s
 => [3/4] COPY . .                                                                                              0.1s
 => [4/4] RUN pip install --no-cache-dir -r requirements.txt                                                   13.7s
 => exporting to image                                                                                          2.8s
 => => exporting layers                                                                                         1.6s
 => => exporting manifest sha256:ccaad06adad139c3587a3b2e079641dc071c79dd6aa4f988e316832c59930798               0.0s
 => => exporting config sha256:f92a204f9a95b24524cf9b05e6ccac9508d4b595d2e2eb94e5e7a66d04505113                 0.0s
 => => exporting attestation manifest sha256:f8e16f95990989d70beafd2a70045895524268826b963f1cd9f1bbe18ff0ea4b    0.1s
 => => exporting manifest list sha256:ab5d64850cad089ae916b35aaeabfed4e2e79ccee41f3a1a524e1f22cb6e586b           0.0s
 => => naming to docker.io/alejoro21/sre-calculator:1.0                                                         0.0s
 => => unpacking to docker.io/alejoro21/sre-calculator:1.0                                                      0.9s
Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

Then pushed to Docker:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ docker push alejoro21/sre-calculator:1.0
The push refers to repository [docker.io/alejoro21/sre-calculator]
22b63e76fde1: Pushed
1771569cc129: Pushed
8b98dd5099f3: Pushed
4da6efc9e61a: Pushed
b3dd773c3296: Pushed
0e4bc2bd6656: Pushed
7ff9f5290b62: Pushed
f27720ea7a18: Pushed
1.0: digest: sha256:ab5d64850cad089ae916b35aaeabfed4e2e79ccee41f3a1a524e1f22cb6e586b size: 856
Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

Then moved directory to K8s to create the yaml file:

```
PROBLEMS 3    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS 2

Joro_21@L-Mejia21:~/projects/sre-final-project/app$ cd ~/projects/sre-final-project/
Joro_21@L-Mejia21:~/projects/sre-final-project$ cd k8s/
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ nano app.yaml
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ ls
app.yaml
```

```
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ cat app.yaml
# Namespace
apiVersion: v1
kind: Namespace
metadata:
  name: monitoring
---
# Deployment de la app
apiVersion: apps/v1
kind: Deployment
metadata:
  name: fastapi-calculator
  namespace: monitoring
  labels:
    app: fastapi-calculator
spec:
  replicas: 2
  selector:
    matchLabels:
      app: fastapi-calculator
  template:
    metadata:
      labels:
        app: fastapi-calculator
    spec:
      containers:
        - name: fastapi
          image: alejoro21/sre-calculator:1.0
          ports:
            - containerPort: 8000
          readinessProbe:
            httpGet:
              path: /
              port: 8000
            initialDelaySeconds: 5
            periodSeconds: 5
          livenessProbe:
            httpGet:
              path: /
              port: 8000
            initialDelaySeconds: 10
```

```
            initialDelaySeconds: 10
            periodSeconds: 10
---
# Service tipo NodePort para exponer API fuera del cluster
apiVersion: v1
kind: Service
metadata:
  name: fastapi-service
  namespace: monitoring
spec:
  type: NodePort
  selector:
    app: fastapi-calculator
  ports:
    - port: 8000        # puerto dentro del cluster
      targetPort: 8000  # puerto del contenedor
      nodePort: 30080   # puerto del nodo (Minikube)
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$
```

Then checked minikube status to ensure it was running and working, fortunately I did, because it wasn't:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ minikube status
minikube
type: Control Plane
host: Stopped
kubelet: Stopped
apiserver: Stopped
kubeconfig: Stopped

Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ minikube start --driver=docker
😄 minikube v1.37.0 on Ubuntu 22.04 (kvm/amd64)
✨ Using the docker driver based on existing profile

❗ The requested memory allocation of 3072MiB does not leave room for system overhead (total system memory: 3534MiB).
You may face stability issues.
💡 Suggestion: Start minikube with less memory allocated: 'minikube start --memory=3072mb'

👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚜 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🐳 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎 Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🏄 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$
```

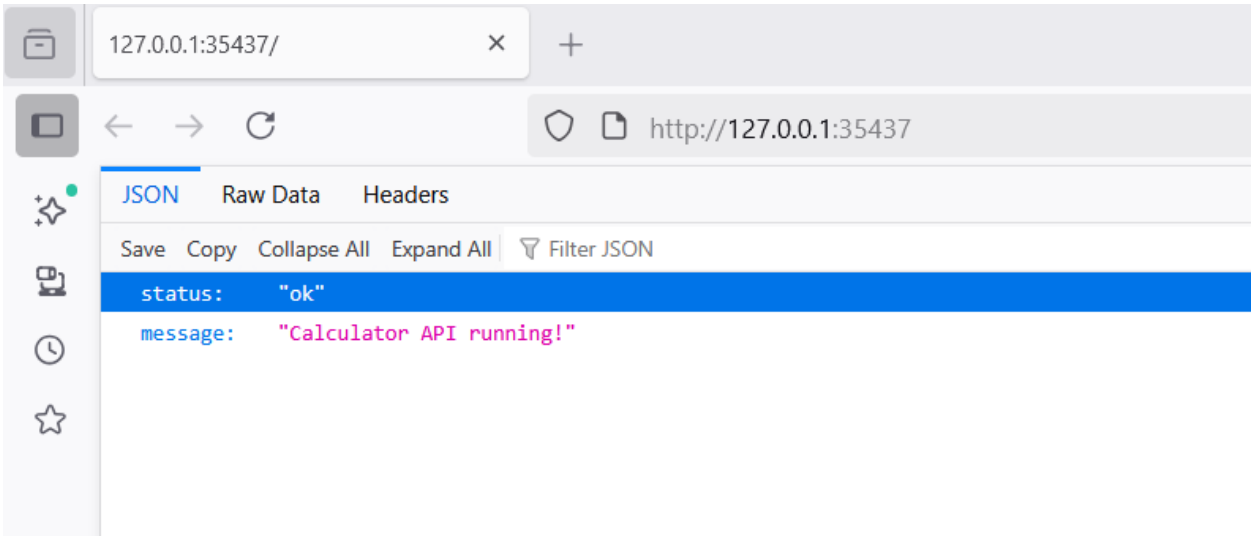
Then applied my Yaml and verified the Pods:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl apply -f app.yaml
namespace/monitoring unchanged
deployment.apps/fastapi-calculator created
service/fastapi-service created
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl get pods -n monitoring
NAME                                  READY   STATUS              RESTARTS        AGE
fastapi-calculator-7854d4586-7r8zt    0/1     ContainerCreating   0               25s
fastapi-calculator-7854d4586-hwqn6    0/1     ContainerCreating   0               25s
fastapi-calculator-7854d4586-pnhqs    0/1     ContainerCreating   0               25s
fastapi-calculator-7854d4586-zvl8w    0/1     ContainerCreating   0               25s
grafana-deployment-6fd749d778-w4clk   1/1     Running             1 (4m42s ago)   12d
prometheus-deployment-597cbfc968-sbf2f   1/1     Running             1 (4m42s ago)   12d
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl get pods -n monitoring
NAME                                  READY   STATUS    RESTARTS        AGE
fastapi-calculator-7854d4586-7r8zt    1/1     Running   0               49s
fastapi-calculator-7854d4586-hwqn6    1/1     Running   0               49s
fastapi-calculator-7854d4586-pnhqs    1/1     Running   0               49s
fastapi-calculator-7854d4586-zvl8w    1/1     Running   0               49s
grafana-deployment-6fd749d778-w4clk   1/1     Running   1 (5m6s ago)    12d
prometheus-deployment-597cbfc968-sbf2f   1/1     Running   1 (5m6s ago)    12d
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$
```

Then deployed the service and reviewed results in web:

```
! app.yaml 1 ×
k8s > ! app.yaml > {} spec > # replicas
8   apiVersion: apps/v1
9   kind: Deployment
10  metadata:
11    name: fastapi-calculator
12    namespace: monitoring
13    labels:
14      app: fastapi-calculator
15  spec:
16    replicas: 4
17    selector:
18      matchLabels:
19        app: fastapi-calculator
20    template:
21      metadata:
```

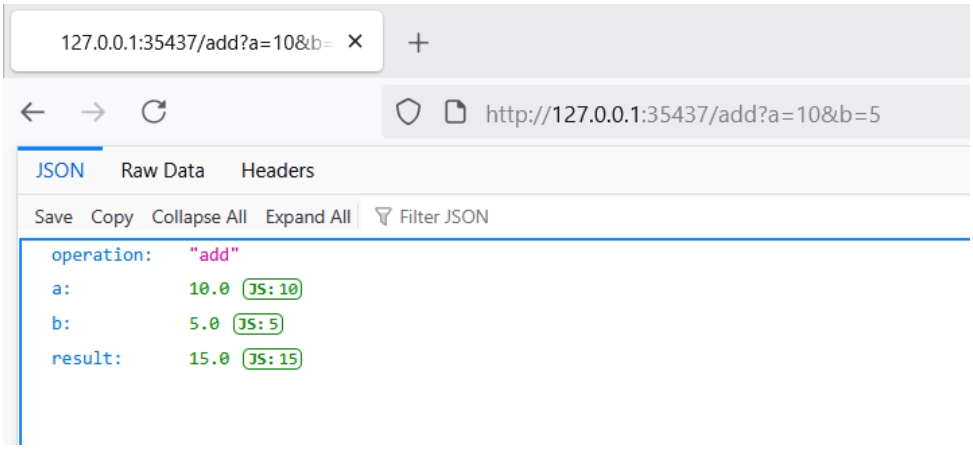
```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS 3          minikube - k8s
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ minikube service fastapi-service -n monitoring --url
! Executing "docker container inspect minikube --format={{.State.Status}}" took an unusually long time: 2.354684298s
💡 Restarting the docker service may improve performance.
http://127.0.0.1:35437
! Because you are using a Docker driver on linux, the terminal needs to be open to run it.
```

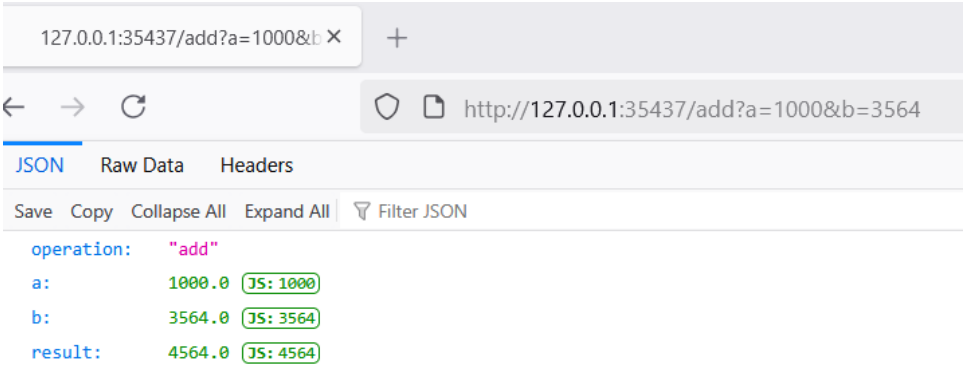


Important fail check in case anything goes wrong:

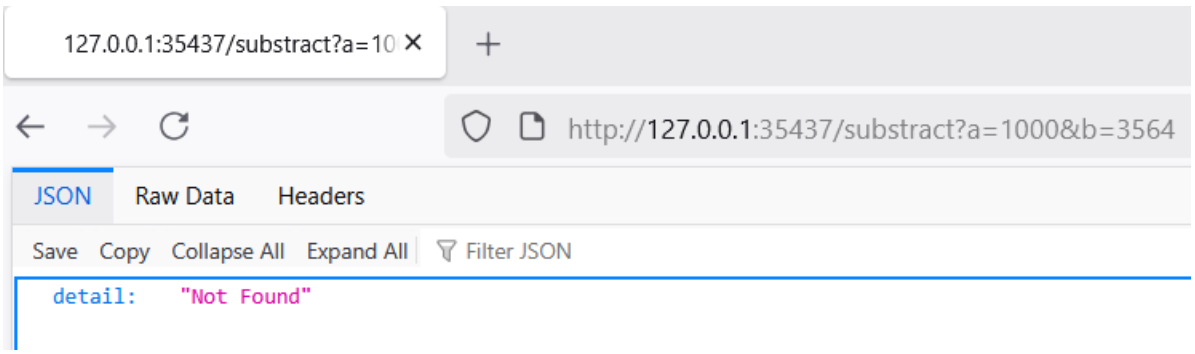
- kubectl get pods -n monitoring → ¿Are Pods Running?
- kubectl describe pod … → Check image or permissions errors
- kubectl logs deploy/fastapi-calculator -n monitoring → Ensure FastAPI did liftoff.
- docker ps in Ubuntu → confirm Docker is OK (if permission denied: newgrp docker)

Ran a couple tests with the app to ensure it worked fine:

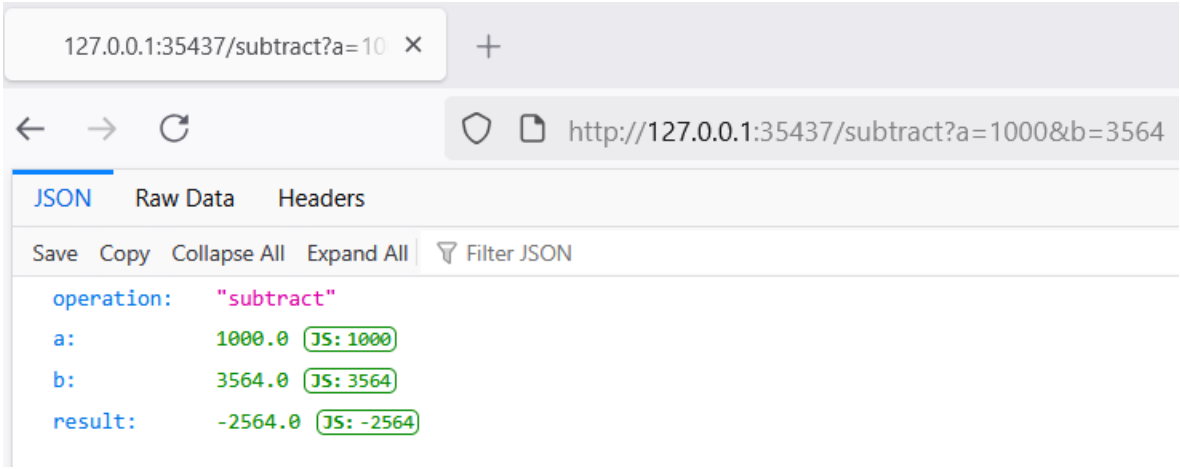


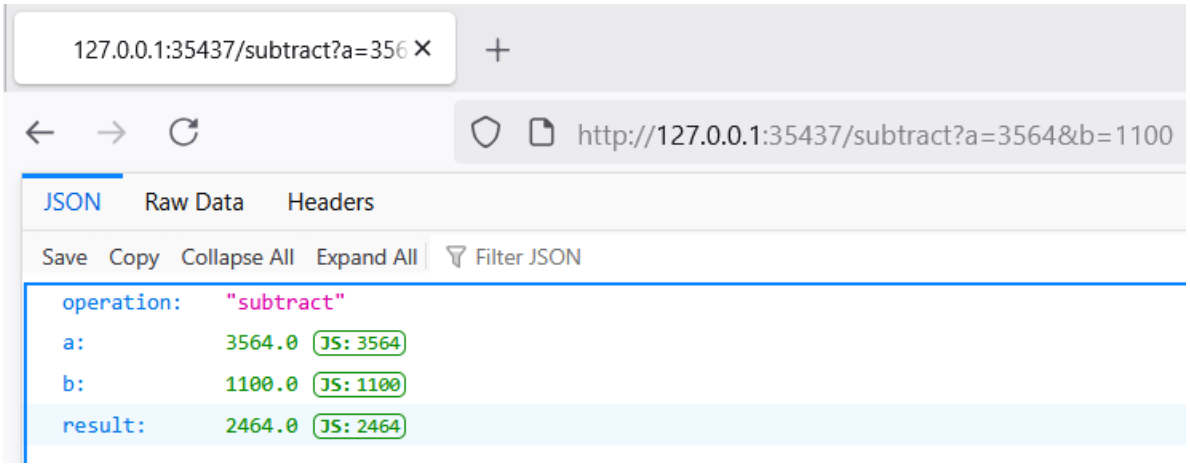


Caused error to review later: Bad name: susbstract

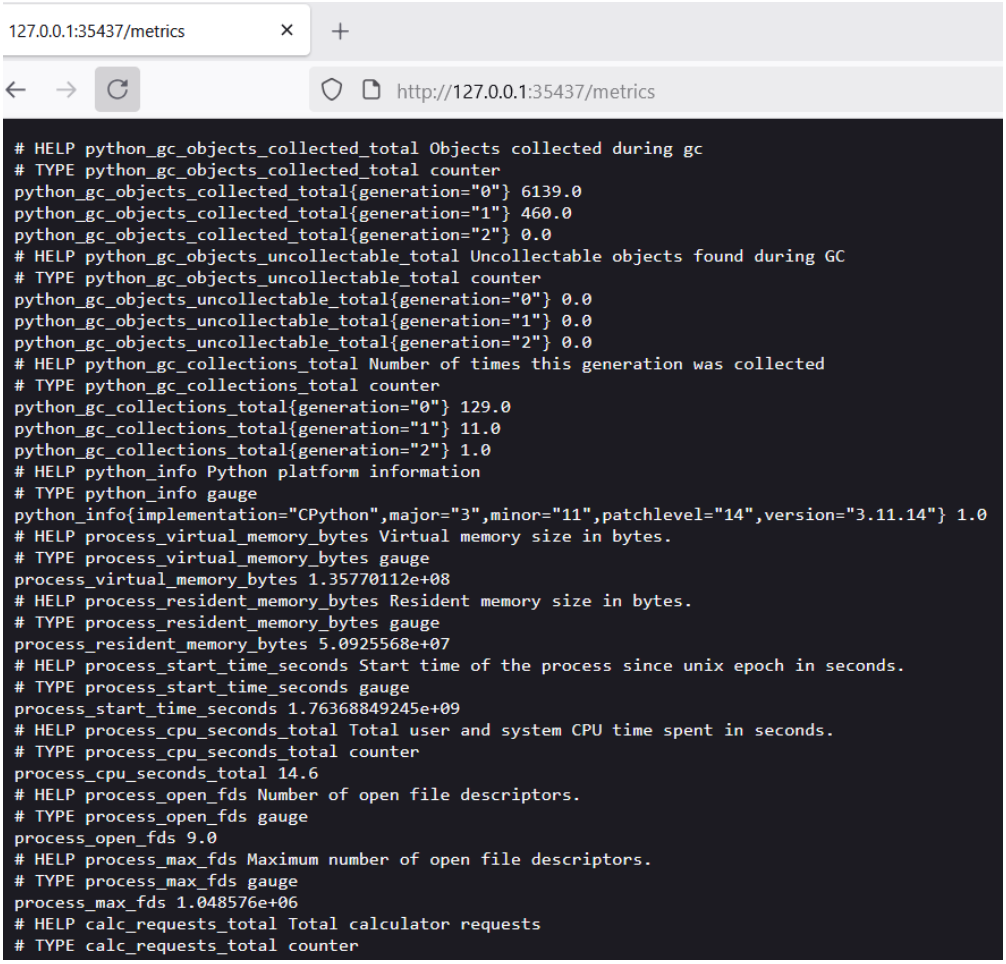


Tested correct:



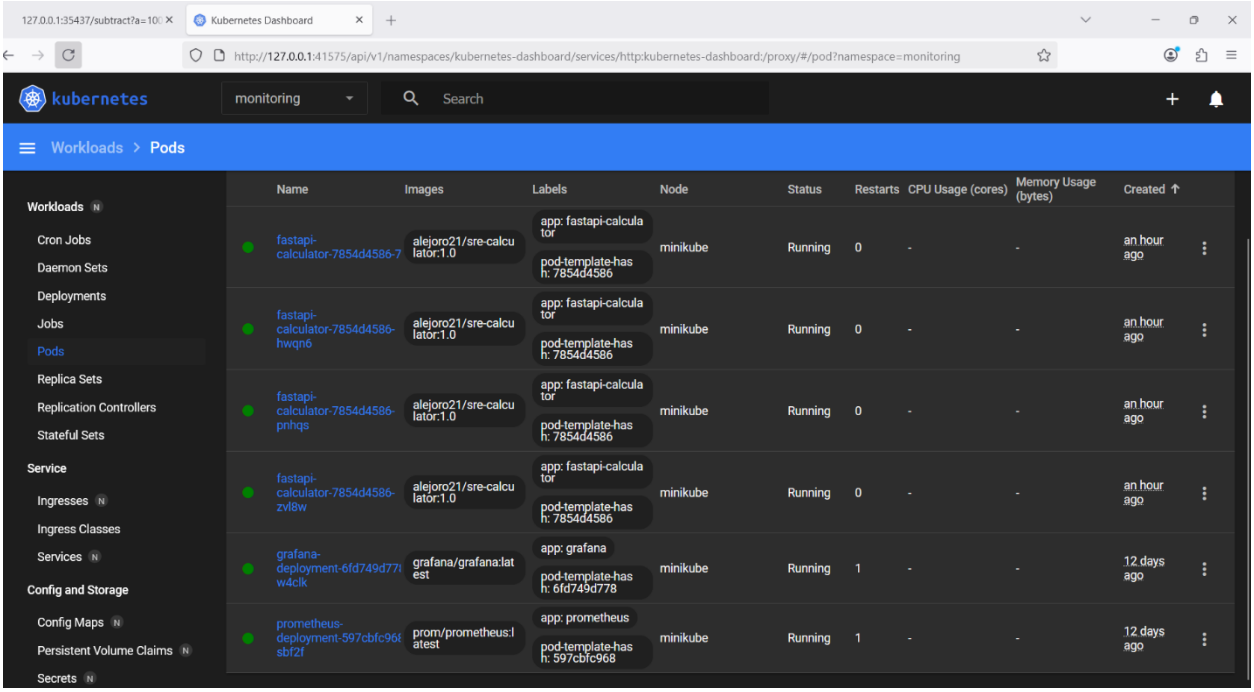


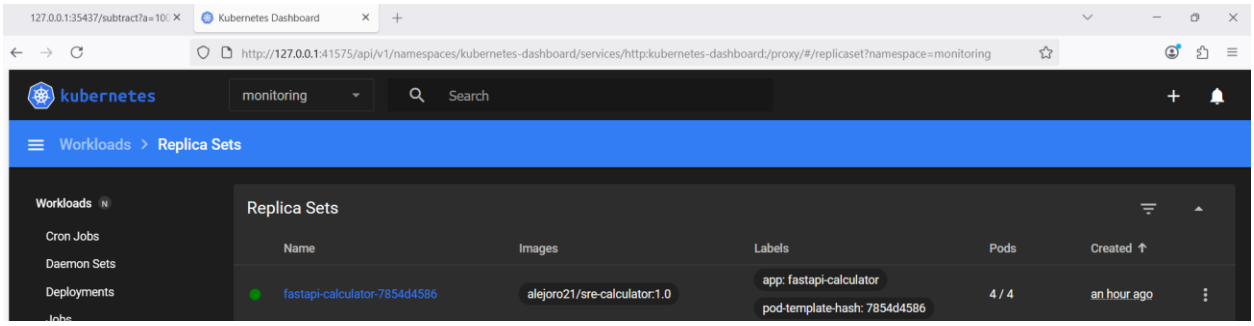
Then got some metrics (raw)




```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 6139.0
python_gc_objects_collected_total{generation="1"} 460.0
python_gc_objects_collected_total{generation="2"} 0.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 129.0
python_gc_collections_total{generation="1"} 11.0
python_gc_collections_total{generation="2"} 1.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="11",patchlevel="14",version="3.11.14"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 1.35770112e+08
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 5.0925568e+07
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.76368849245e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 14.6
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 9.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 1.048576e+06
# HELP calc_requests_total Total calculator requests
# TYPE calc_requests_total counter
```

Checked on the Kubernetes Dashboard just for the nice resource view:





Reviewed pods and service one more time:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl get pods -n monitoring
NAME                                    READY   STATUS    RESTARTS      AGE
fastapi-calculator-7854d4586-7r8zt      1/1     Running   0             75m
fastapi-calculator-7854d4586-hwqn6      1/1     Running   0             75m
fastapi-calculator-7854d4586-pnhqs      1/1     Running   0             75m
fastapi-calculator-7854d4586-zvl8w      1/1     Running   0             75m
grafana-deployment-6fd749d778-w4clk     1/1     Running   1 (80m ago)   12d
prometheus-deployment-597cbfc968-sbf2f  1/1     Running   1 (80m ago)   12d
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl get svc -n monitoring
NAME                 TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
fastapi-service      NodePort   10.106.90.111    <none>        8000:30080/TCP   76m
grafana-service      NodePort   10.108.226.159   <none>        3000:30632/TCP   12d
prometheus-service   NodePort   10.107.198.131   <none>        9090:32664/TCP   12d
```

So far, I've been able to:

- Create and test my own functional calculator in FastAPI.
- Containerized in Docker.
- Imaged pushed with Docker Hub.
- Deployed in Kubernetes with 4 replicas.
- Exposed to the world via NodePort/service.
- Endpoint /metrics ready for Prometheus.

Now, will be moving forward to Prometheus and Grafana.

Mentor mentioned it is not necessary to have the Prometheus.yaml local in the k8s folder in order for it to work.

Entered vi version of Prometheus yaml and added the extra block in scrape configs:

```
- job_name: 'fastapi-calculator'
    static_configs:
      - targets: ['fastapi-service.monitoring.svc.cluster.local:8000']
```

```
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ ls
app.yaml
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl get pods -n monitoring
NAME                                    READY   STATUS    RESTARTS      AGE
fastapi-calculator-67b8964b5f-6g58b     1/1     Running   0             56m
fastapi-calculator-67b8964b5f-csrhn     1/1     Running   0             56m
fastapi-calculator-67b8964b5f-n6hcw     1/1     Running   0             56m
fastapi-calculator-67b8964b5f-wfmlb     1/1     Running   0             56m
grafana-deployment-6fd749d778-w4clk     1/1     Running   1 (3h ago)    12d
prometheus-deployment-597cbfc968-sbf2f  1/1     Running   1 (3h ago)    12d
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl get svc -n monitoring
NAME                 TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
fastapi-service      NodePort   10.106.90.111    <none>        8000:30080/TCP   177m
grafana-service      NodePort   10.108.226.159   <none>        3000:30632/TCP   12d
prometheus-service   NodePort   10.107.198.131   <none>        9090:32664/TCP   12d
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl -n monitoring edit configmap prometheus-config
configmap/prometheus-config edited
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$
```

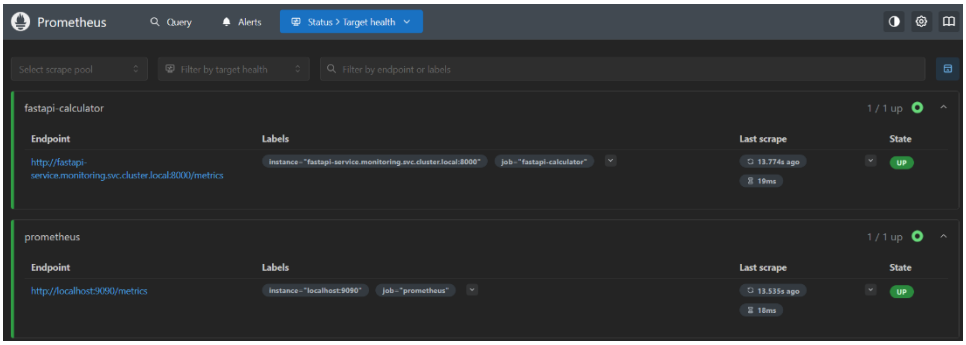
Restarted Prometheus for it to get the new config:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl -n monitoring edit configmap prometheus-config
configmap/prometheus-config edited
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl -n monitoring rollout restart deploy/prometheus-deployment
deployment.apps/prometheus-deployment restarted
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl get pods -n monitoring
NAME                                    READY   STATUS    RESTARTS        AGE
fastapi-calculator-67b8964b5f-6g58b     1/1     Running   0               77m
fastapi-calculator-67b8964b5f-csrhn     1/1     Running   0               77m
fastapi-calculator-67b8964b5f-n6hcw     1/1     Running   0               77m
fastapi-calculator-67b8964b5f-wfmlb     1/1     Running   0               77m
grafana-deployment-6fd749d778-w4clk     1/1     Running   1 (3h22m ago)   12d
prometheus-deployment-5c4f87cf94-h465r  1/1     Running   0               20s
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$
```

Then checked Prometheus status:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ minikube service prometheus-service -n monitoring --url
❗  Executing "docker container inspect minikube --format={{.State.Status}}" took an unusually long time: 4.482963143s
💡  Restarting the docker service may improve performance.
http://127.0.0.1:36155
❗  Because you are using a Docker driver on linux, the terminal needs to be open to run it.
```

Note: Prometheus config used in exercises has a Cameo here:



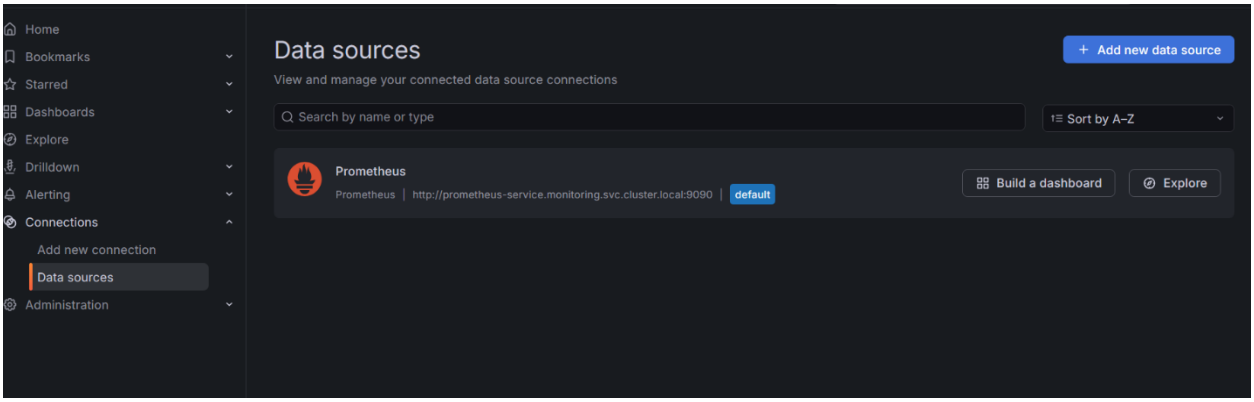
Moving to Grafana with Prometheus up and running!

Opened Grafana:

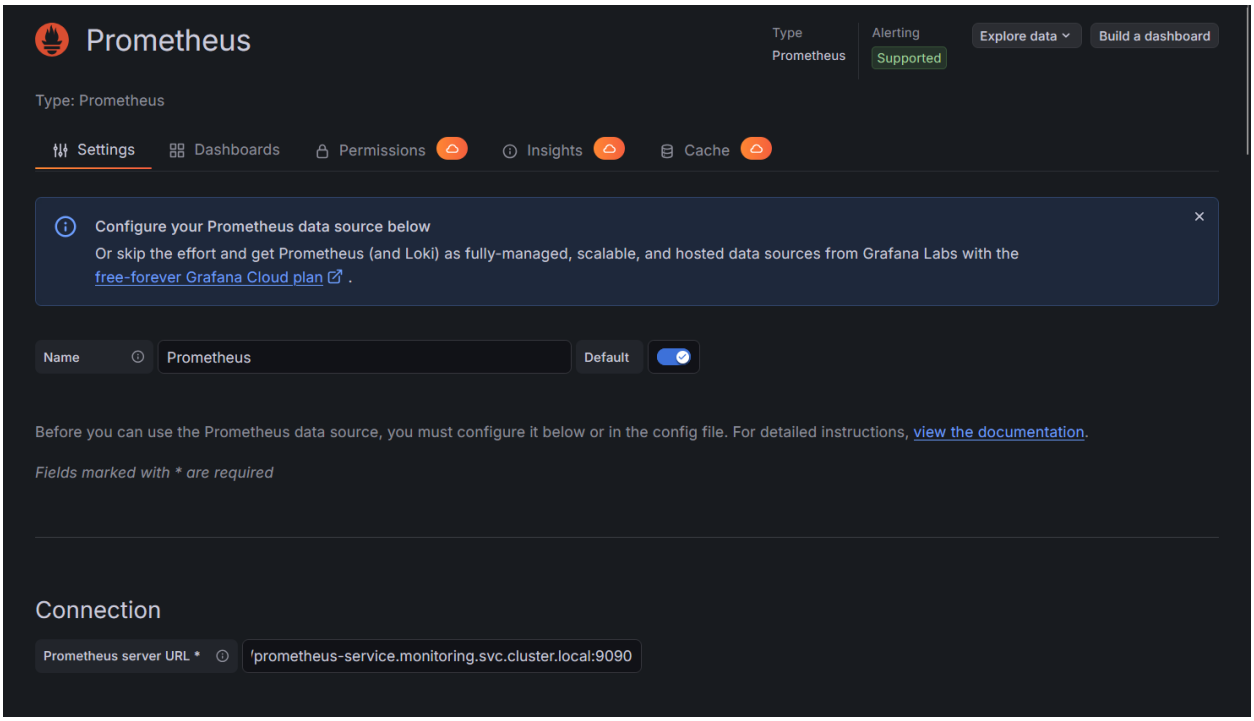
```
Joro_21@L-Mejia21:~/projects/sre-final-project$ minikube service grafana-service -n monitoring --url
http://127.0.0.1:39781
  ❗  Because you are using a Docker driver on linux, the terminal needs to be open to run it.
```

Discovered there was an update and data sources are no longer in Administration tab, but in **connections tab**.

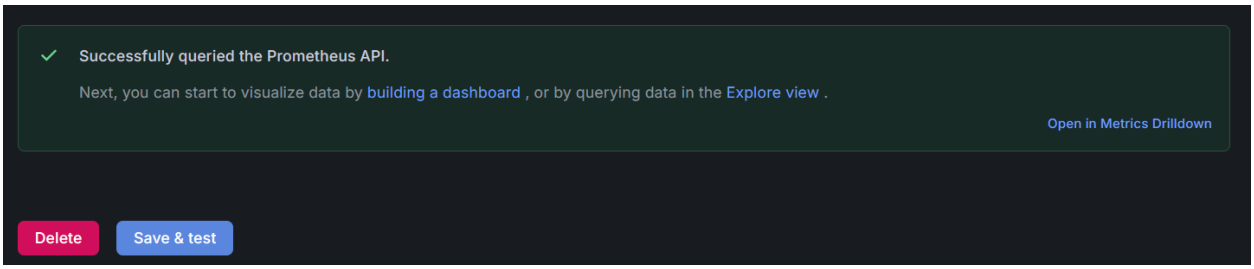
Prometheus running:



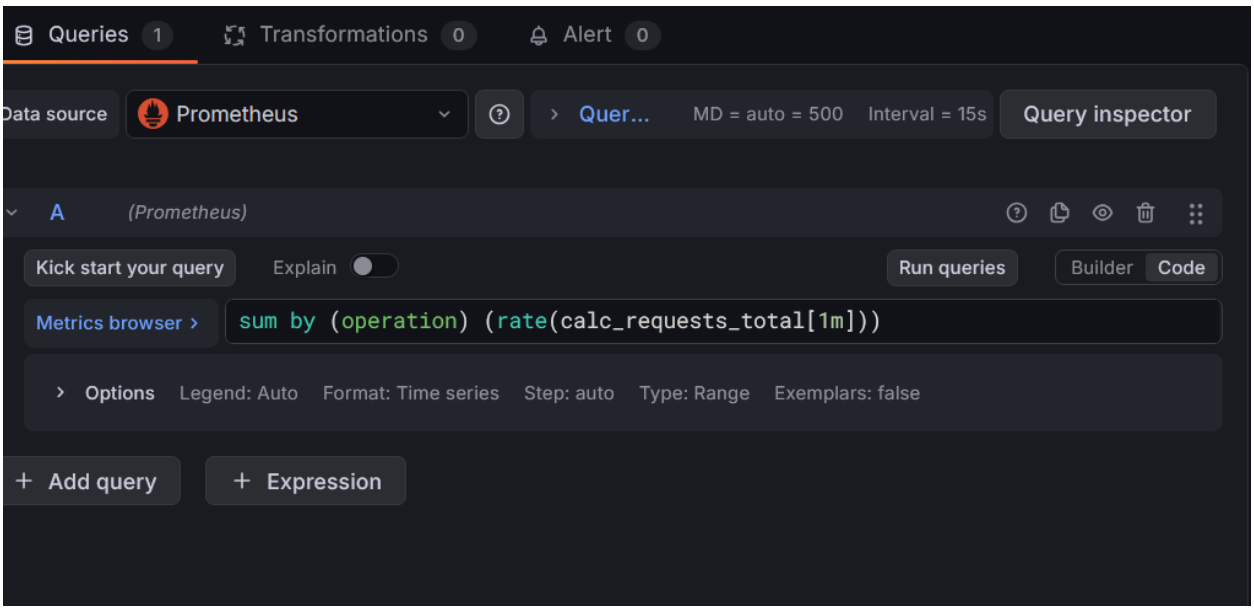
Ensured the link is updated:



Going into dashboard mode!



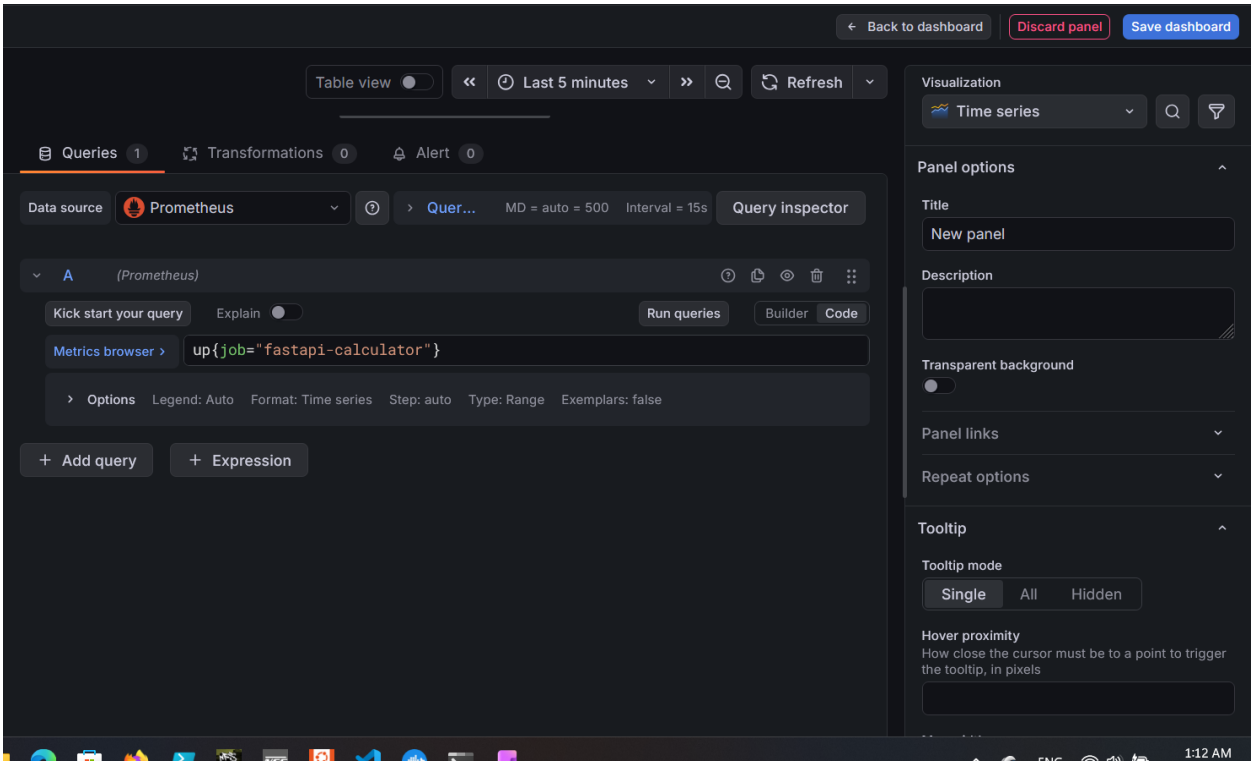
Ran into a little issue about not seeing results or any changes made when entered a query:



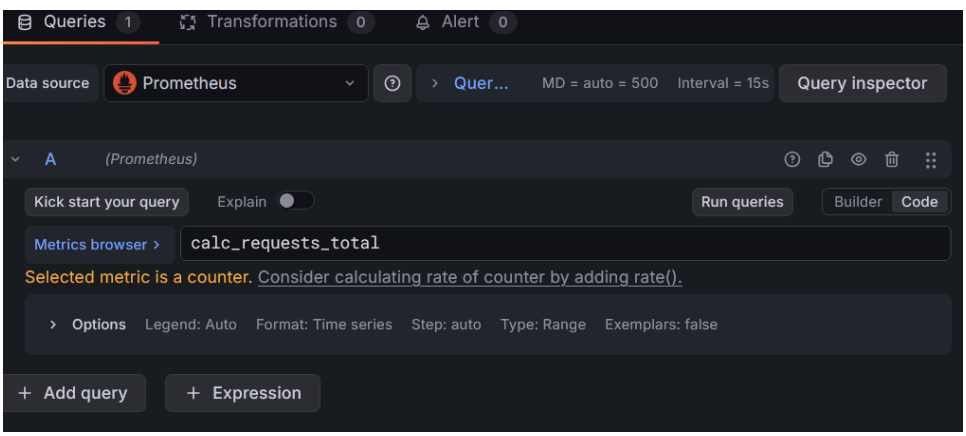
Query: sum by (operation) (rate(calc_requests_total[1m]))

Also tried: up{job="fastapi-calculator"}

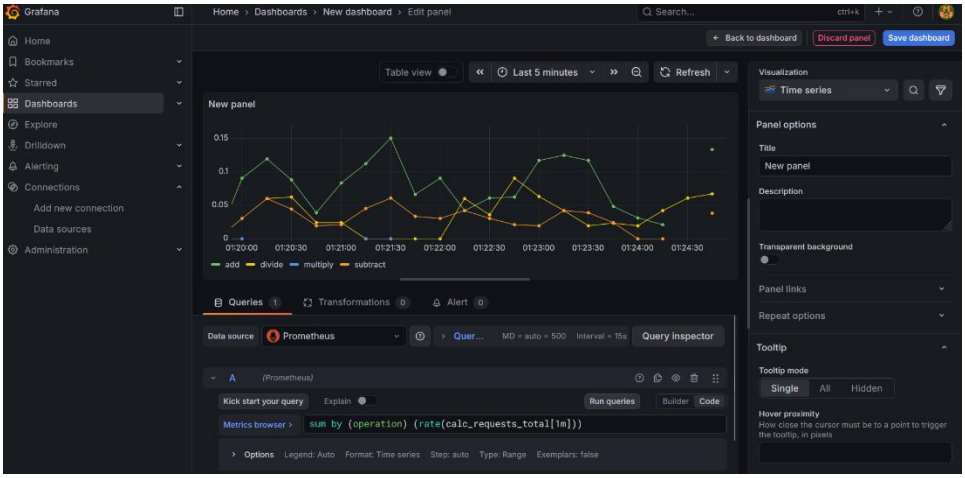
Had to learn the hard way that Prometheus is kind of Shy.



Still no results after testing the calculator several times and waiting for Prometheus to scrap /metrics. Attempted without rate and it gave me an error too:

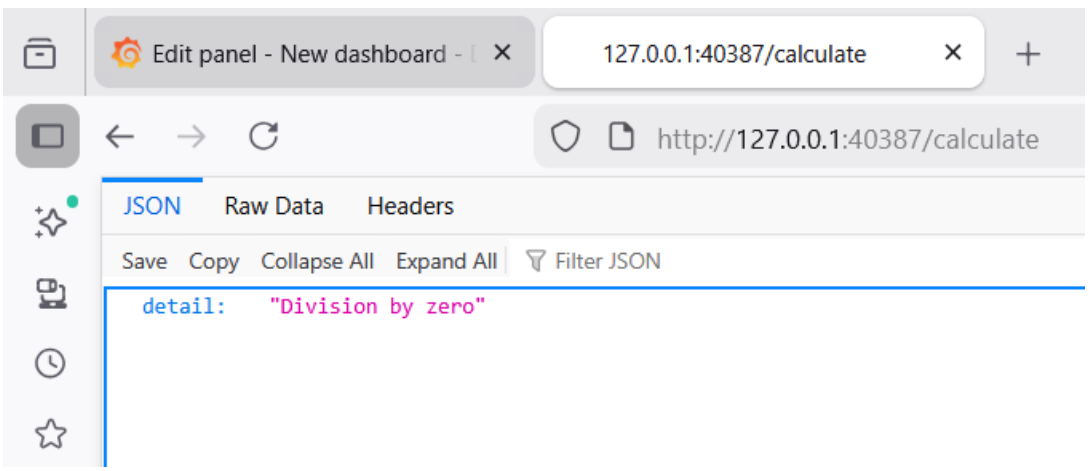


Then realized I had hidden the panel to focus on the query and results were present:



Confirmed Prometheus is scrapping the correct information and Grafana is alert ready!

Generated error for Grafana to pick and review the alert:



Ansible setup:

To setup my automation with Ansible, I started by copying the prometheus and grafana docs from the exercises folders. Then I proceeded to create my inventory.ini file:

```
Joro_21@L-Mejia21:~/projects/sre-final-project$ cp ~/projects/sre-academy-training-main/exercises/exercise5/prometheus.yaml k8s/
Joro_21@L-Mejia21:~/projects/sre-final-project$ cp ~/projects/sre-academy-training-main/exercises/exercise6/grafana.yaml k8s/
Joro_21@L-Mejia21:~/projects/sre-final-project$ cd k8s/
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ ls
app.yaml  grafana.yaml  prometheus.yaml
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ cd ~/projects/sre-final-project/
Joro_21@L-Mejia21:~/projects/sre-final-project$ mkdir -p ansible
Joro_21@L-Mejia21:~/projects/sre-final-project$ cd a
ansible/ app/
Joro_21@L-Mejia21:~/projects/sre-final-project$ cd ansible/
Joro_21@L-Mejia21:~/projects/sre-final-project/ansible$ nano inventory.ini
Joro_21@L-Mejia21:~/projects/sre-final-project/ansible$ cat
in   ^[[3~^[[3~^C
Joro_21@L-Mejia21:~/projects/sre-final-project/ansible$ cat inventory.ini
[local]
127.0.0.1 ansible_connection=local
Joro_21@L-Mejia21:~/projects/sre-final-project/ansible$
```

Then created my deploy_stack.yaml file:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS 10

Joro_21@L-Mejia21:~/projects/sre-final-project/ansible$ nano deploy_stack.yaml
Joro_21@L-Mejia21:~/projects/sre-final-project/ansible$ cat deploy_stack.yaml
- name: Deploy full SRE stack to local Kubernetes (minikube)
  hosts: local
  gather_facts: false

  vars:
    project_root: "{{ playbook_dir }}/.."

  tasks:
    - name: Start minikube (idempotente, si ya está corriendo no pasa nada)
      ansible.builtin.command:
        cmd: minikube start --driver=docker
      register: minikube_start
      changed_when: "'Done!' in minikube_start.stdout or 'Starting' in minikube_start.stdout"

    - name: Apply Prometheus manifests
      ansible.builtin.command:
        cmd: kubectl apply -f k8s/prometheus.yaml
      args:
        chdir: "{{ project_root }}"

    - name: Apply Grafana manifests
      ansible.builtin.command:
        cmd: kubectl apply -f k8s/grafana.yaml
      args:
        chdir: "{{ project_root }}"

    - name: Apply FastAPI calculator app (Deployment + Service)
      ansible.builtin.command:
        cmd: kubectl apply -f k8s/app.yaml
      args:
        chdir: "{{ project_root }}"

    - name: Show pods in monitoring namespace
      ansible.builtin.command:
        cmd: kubectl get pods -n monitoring
      register: pods_output

    - name: Print pods
      ansible.builtin.debug:
        var: pods_output.stdout
Joro_21@L-Mejia21:~/projects/sre-final-project/ansible$
```

Then, testing time came. Closed all the terminals with the running app and started my ansible with the code:

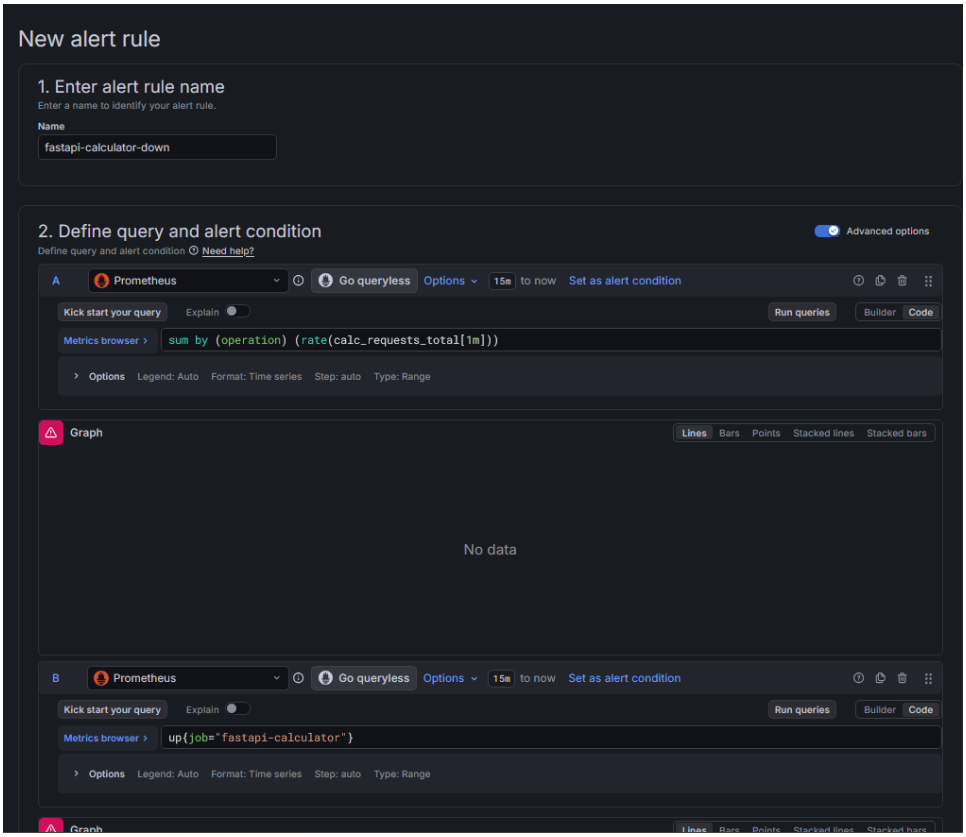
ansible-playbook -i ansible/inventory.ini ansible/deploy_stack.yaml

Took a couple minutes, but it all was running!!!

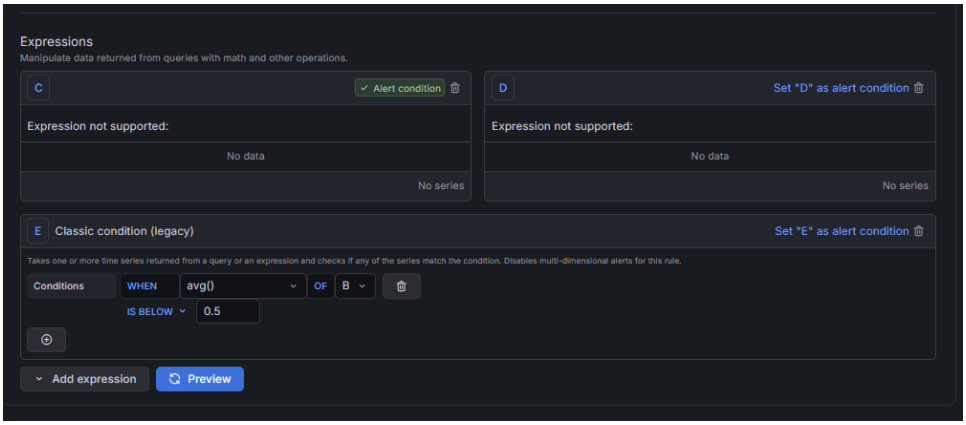
```
Joro_21@L-Mejia21:~/projects/sre-final-project$ ansible-playbook -i ansible/inventory.ini ansible/deploy_stack.yaml

PLAY [Deploy full SRE stack to local Kubernetes (minikube)] ************************************************************************************************************

TASK [Start minikube (idempotente, si ya está corriendo no pasa nada)] ************************************************************************************************
[WARNING]: Platform linux on host 127.0.0.1 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
changed: [127.0.0.1]

TASK [Apply Prometheus manifests] ***************************************************************************************************************************************
changed: [127.0.0.1]

TASK [Apply Grafana manifests] ******************************************************************************************************************************************
changed: [127.0.0.1]

TASK [Apply FastAPI calculator app (Deployment + Service)] **************************************************************************************************************
changed: [127.0.0.1]

TASK [Show pods in monitoring namespace] ********************************************************************************************************************************
changed: [127.0.0.1]

TASK [Print pods] *******************************************************************************************************************************************************
ok: [127.0.0.1] => {
    "pods_output.stdout": "NAME                                      READY   STATUS    RESTARTS      AGE\nfastapi-calculator-747b9bb567-h9clg      1/1     Running   1 (30s ago)   42m\nfastapi
-calculator-747b9bb567-hqq64      1/1     Running   1 (30s ago)   42m\nfastapi-calculator-747b9bb567-w7jt4      1/1     Running   1 (30s ago)   42m\nfastapi-calculator-747b9bb567-wb6bm
1/1     Running   1 (30s ago)   42m\ngrafana-deployment-6fd749d778-w4clk      1/1     Running   5 (33s ago)   14d\nprometheus-deployment-5c4f87cf94-h465r   1/1     Running   4 (32s ago)   47
h"
}

PLAY RECAP **************************************************************************************************************************************************************
127.0.0.1                  : ok=6    changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl get pods -n monitoring
NAME                                      READY   STATUS    RESTARTS        AGE
fastapi-calculator-747b9bb567-h9clg       1/1     Running   1 (3m8s ago)    45m
fastapi-calculator-747b9bb567-hqq64       1/1     Running   1 (3m8s ago)    45m
fastapi-calculator-747b9bb567-w7jt4       1/1     Running   1 (3m8s ago)    45m
fastapi-calculator-747b9bb567-wb6bm       1/1     Running   1 (3m8s ago)    45m
grafana-deployment-6fd749d778-w4clk       1/1     Running   5 (3m11s ago)   14d
prometheus-deployment-5c4f87cf94-h465r    1/1     Running   4 (3m10s ago)   47h
Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl get svc -n monitoring
NAME                 TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
fastapi-service      NodePort   10.106.90.111    <none>        8000:30080/TCP   2d2h
grafana-service      NodePort   10.108.226.159   <none>        3000:30632/TCP   14d
prometheus-service   NodePort   10.107.198.131   <none>        9090:32664/TCP   14d
Joro_21@L-Mejia21:~/projects/sre-final-project$
```

Then, it was time to open everything again in the browser: the app, Prometheus and Grafana to get the monitoring running and looking pretty.

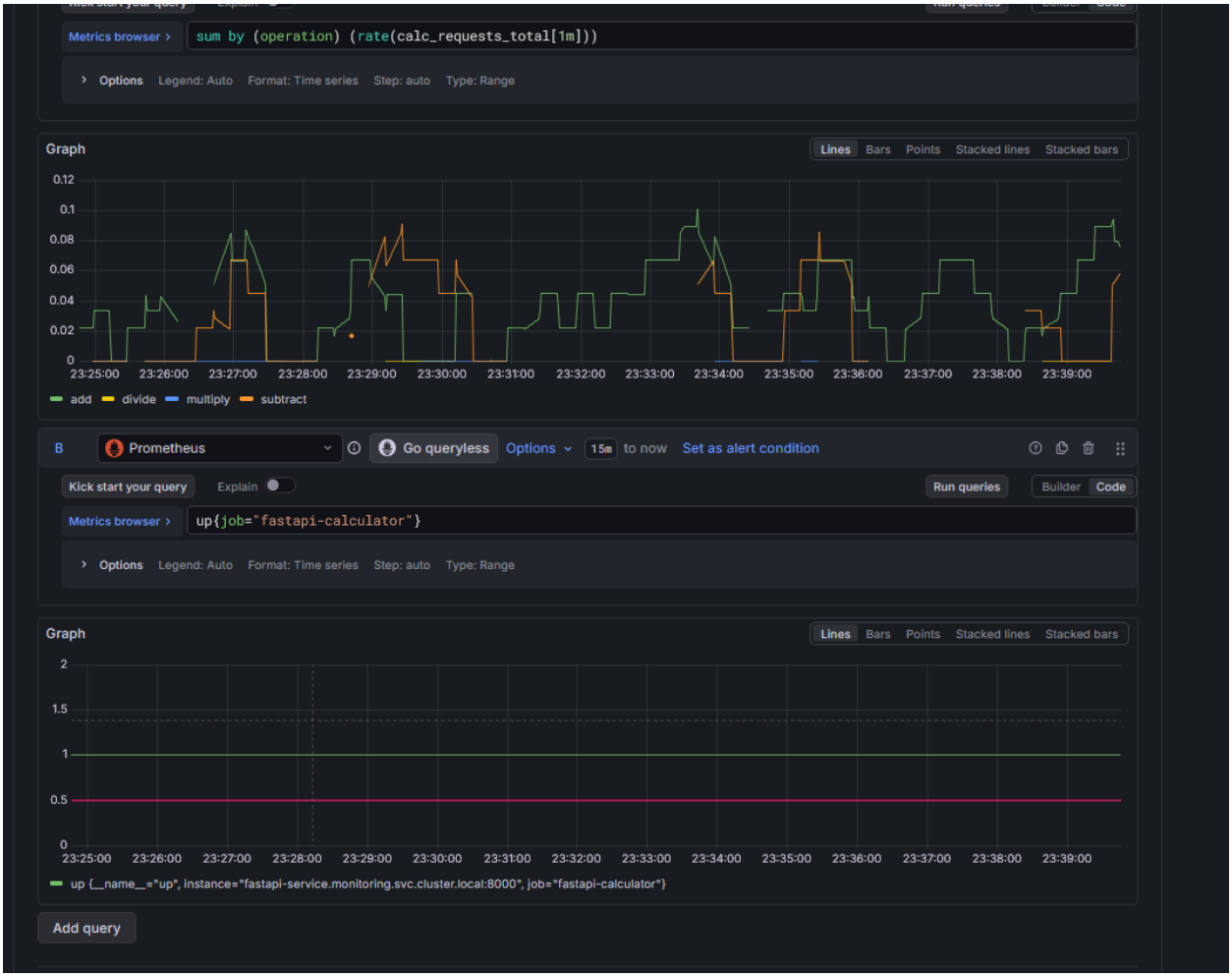
Once everything was running again, started making the steps in Grafana to setup the alerts. Decided to start with a simple alert: app up time. IF below to 0.5 for one minute, an alert will be triggered.



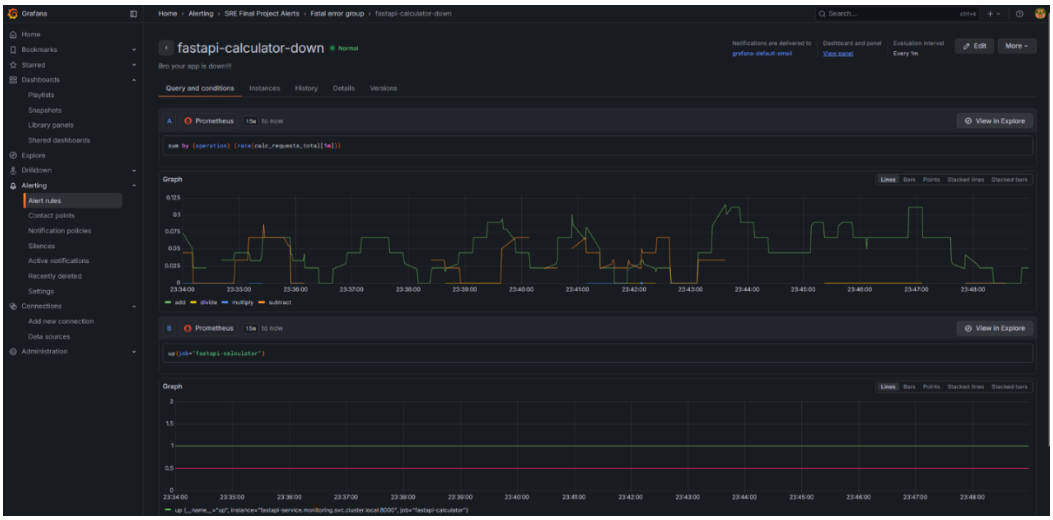
Had some little errors because I added a new expression, instead of using C. No data was prompting.



Deleted C and D, left only E as a condition (expression) and it showed data:



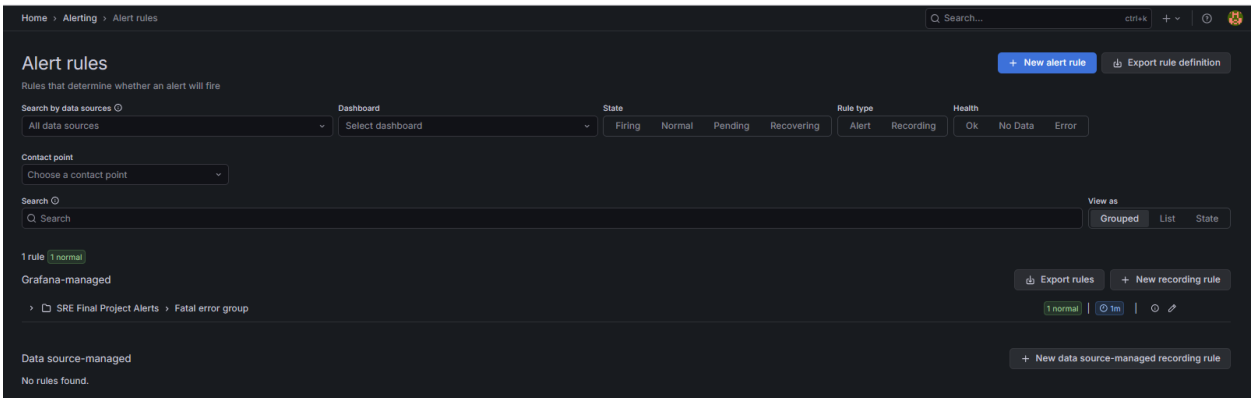
Rule was created, testing with personal email:



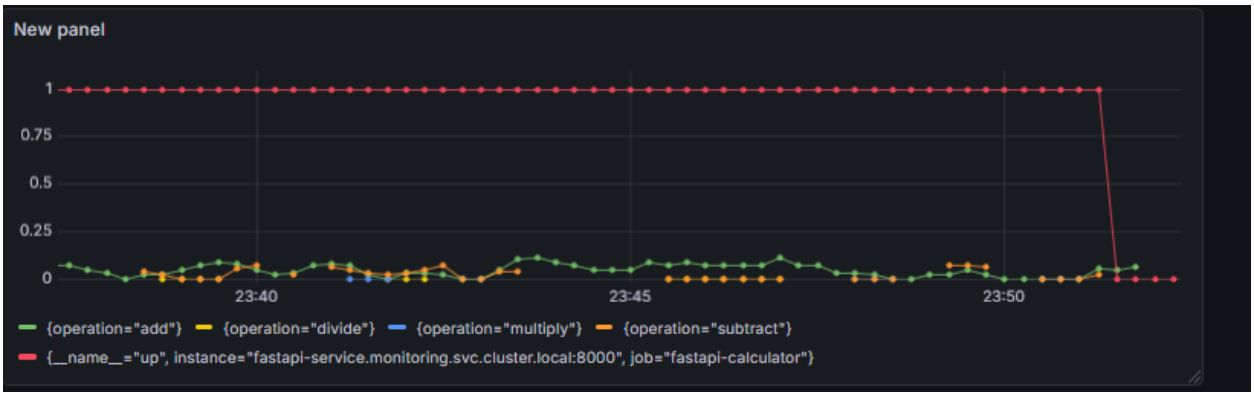
Tested app fall down with code:

kubectl scale deployment fastapi-calculator -n monitoring --replicas=0

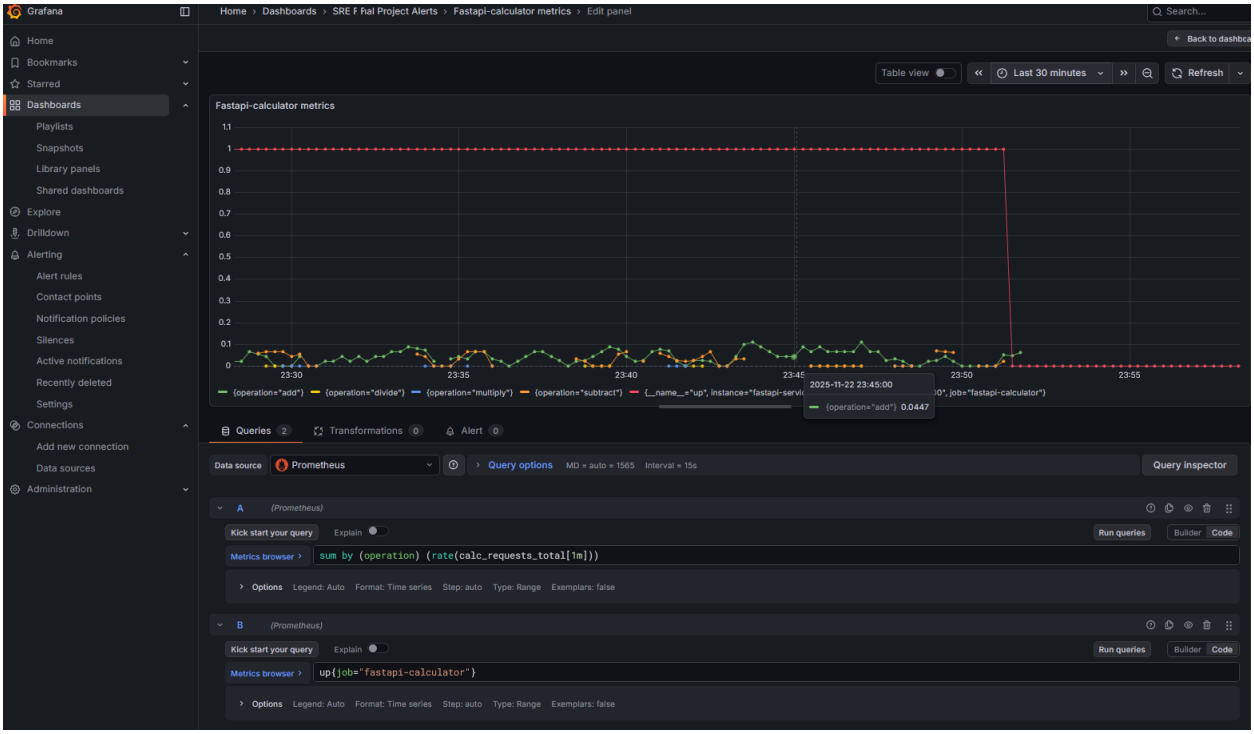
First check before 1 minute passed: (I would later understand my error when I noticed the name of the dashboard the alerts are linked to. Fixed it for next step).



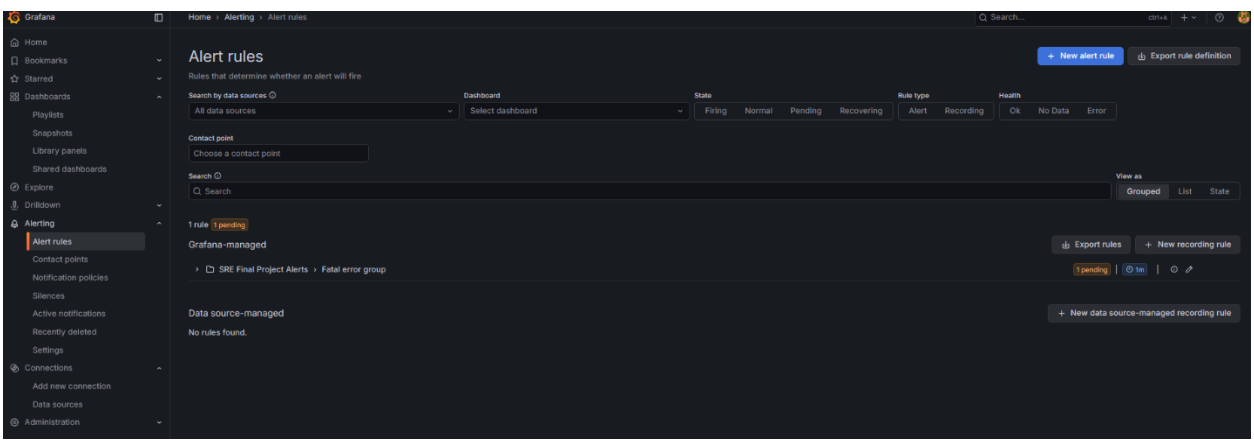
Dashboard does report fall of the app:



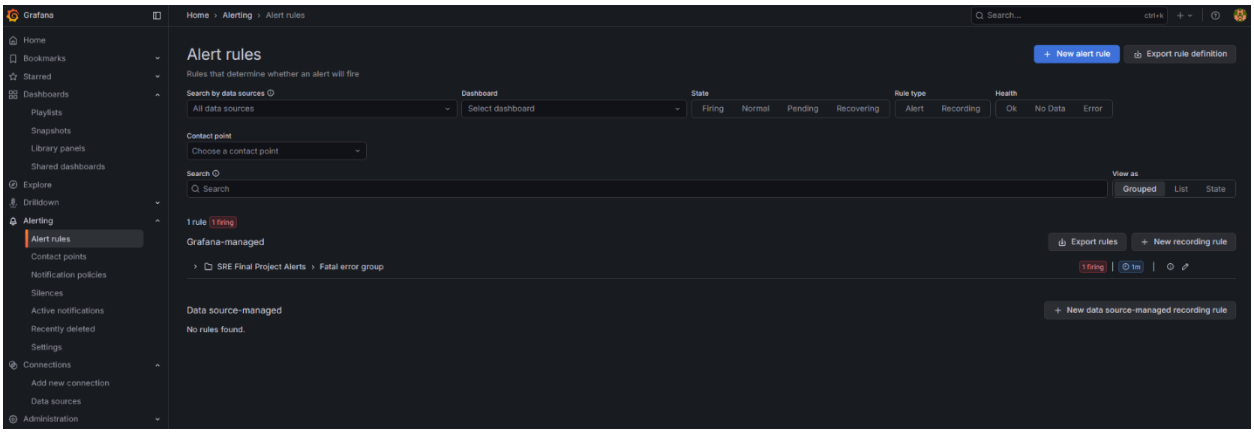
Fixed dashboard for alert:



Alert then changed to pending:

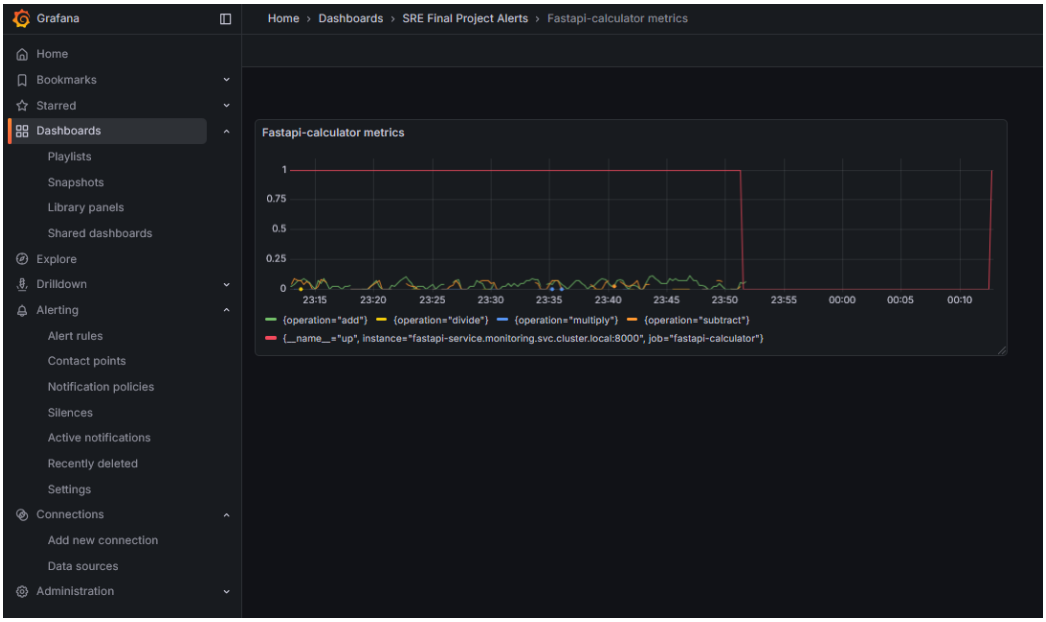


Next minute it updated to firing:

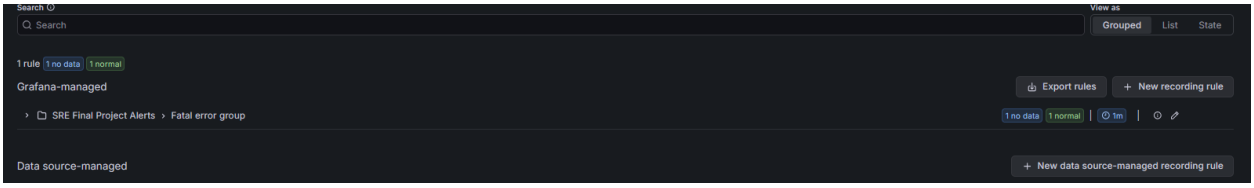


Alert was NOT sent though it fired. I figured the Grafana is configured right (the alert and dashboard), but then had to research a bit. Brought back the app to have it running, Grafana did take the change:

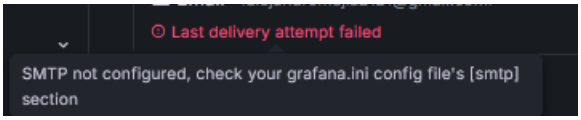
Code used to bring the app back up: kubectl scale deployment fastapi-calculator -n monitoring --replicas=4



Alert went back to normal:

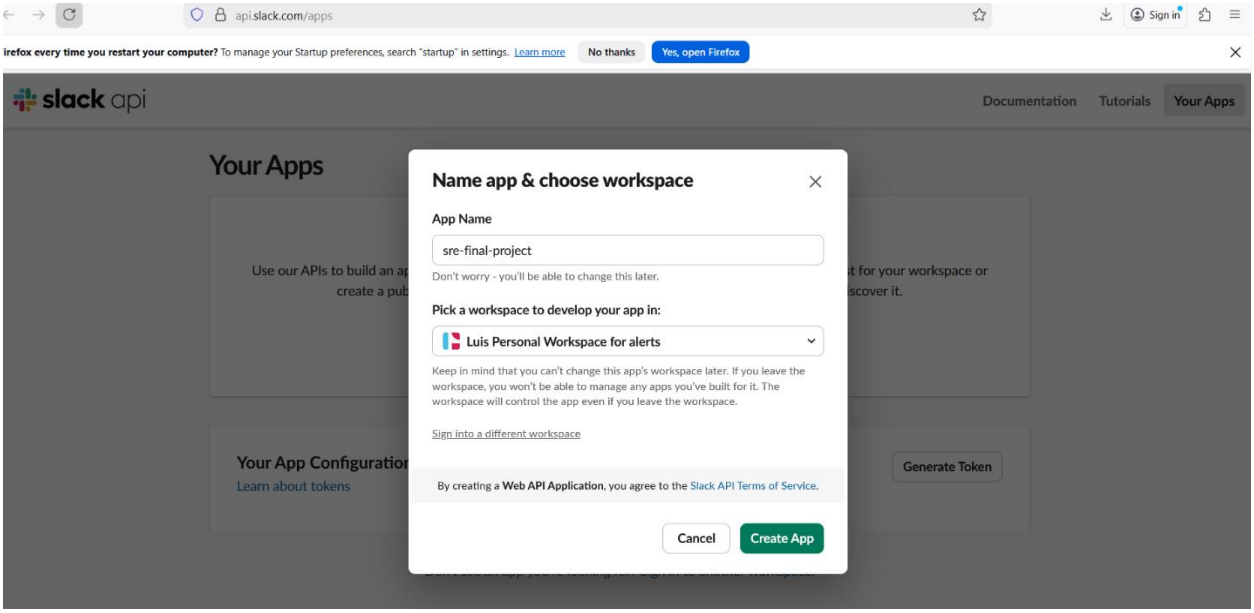


Fortunately, Grafana gave me the error real quick:

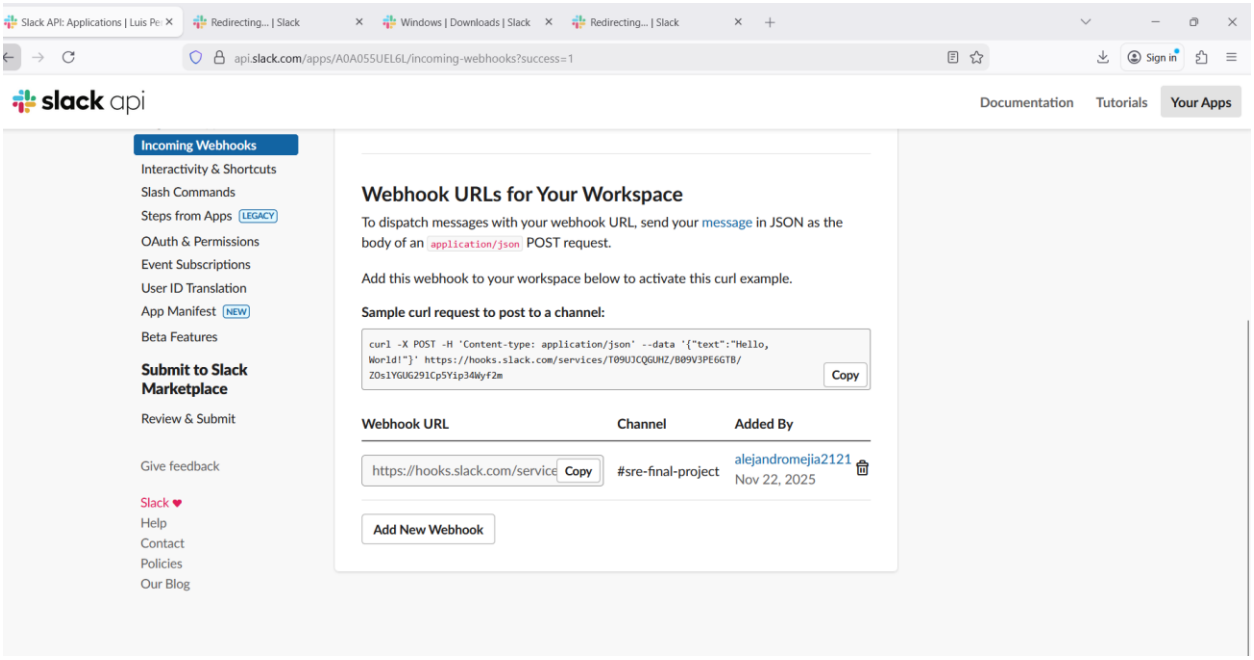


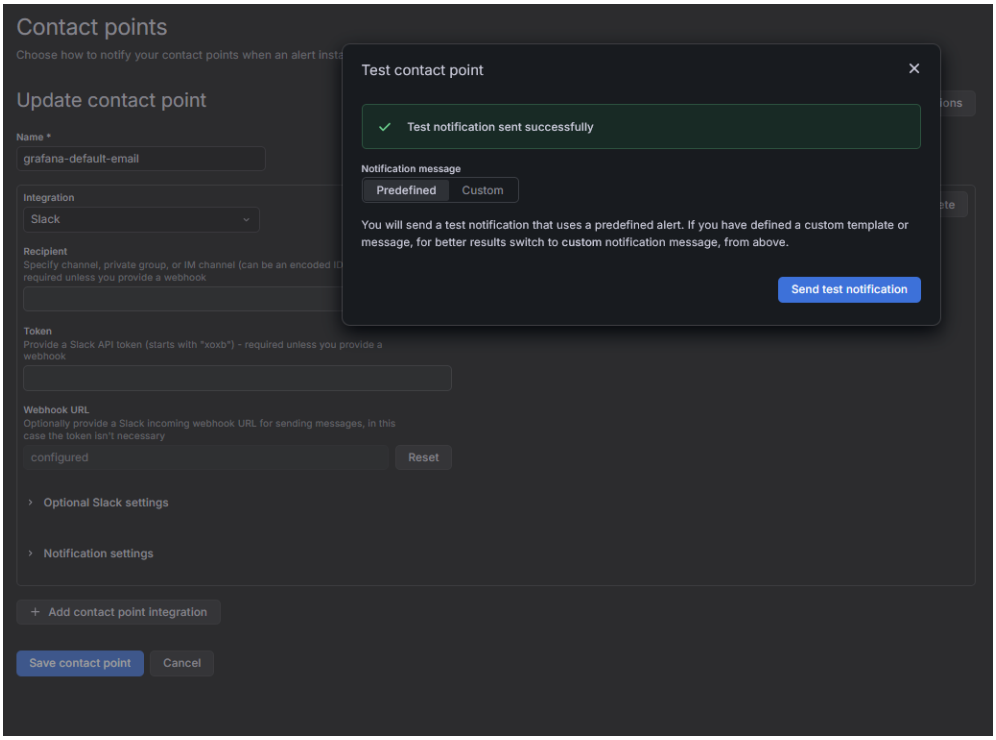
Since SMTP config is a bit more complicated, I made a channel in Slack and got the URL for the Alert to be sent and not only triggered:

Had to create my workspace and an app in slack:

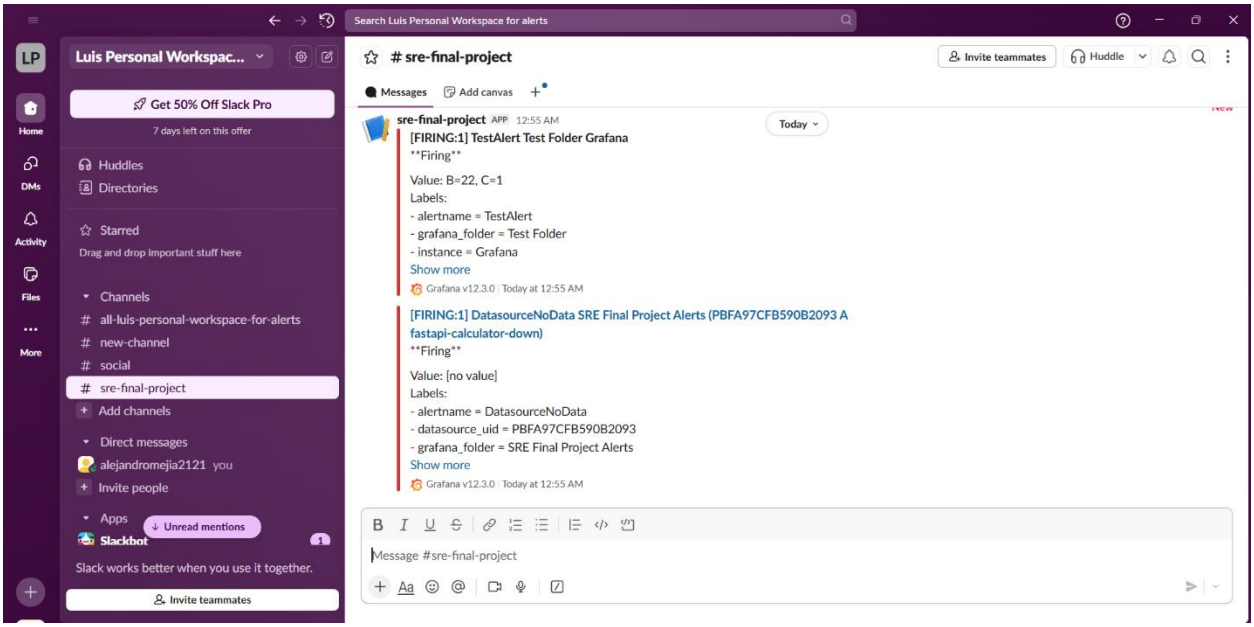


Obtained my webhook link and then moved to Grafana again to set it up:

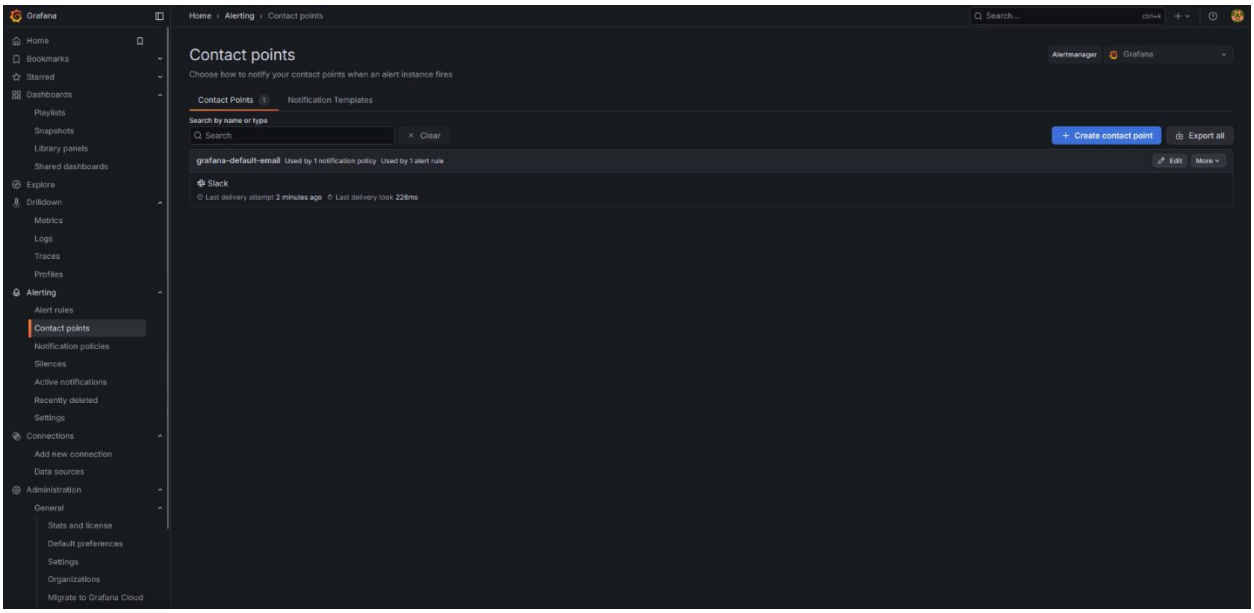




Got my first real notification from Grafana!! But it was just a test:



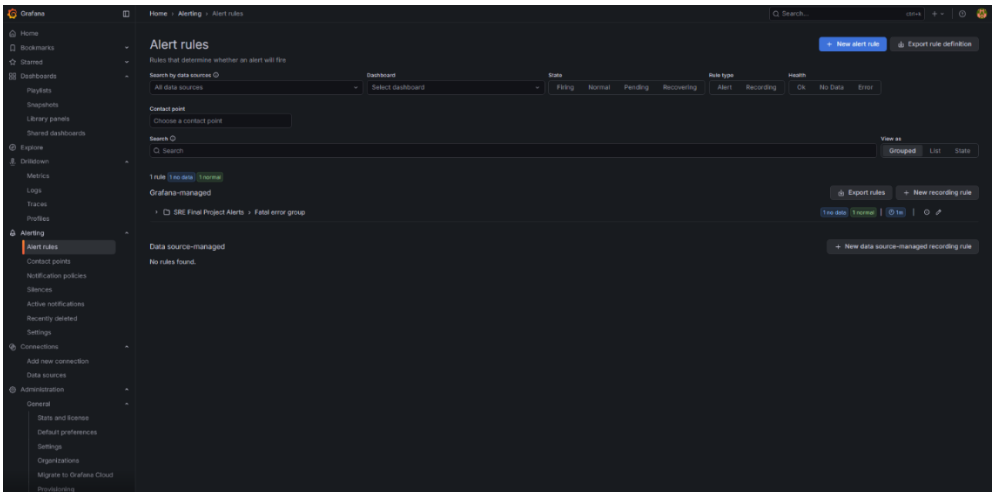
Looks in Grafana:



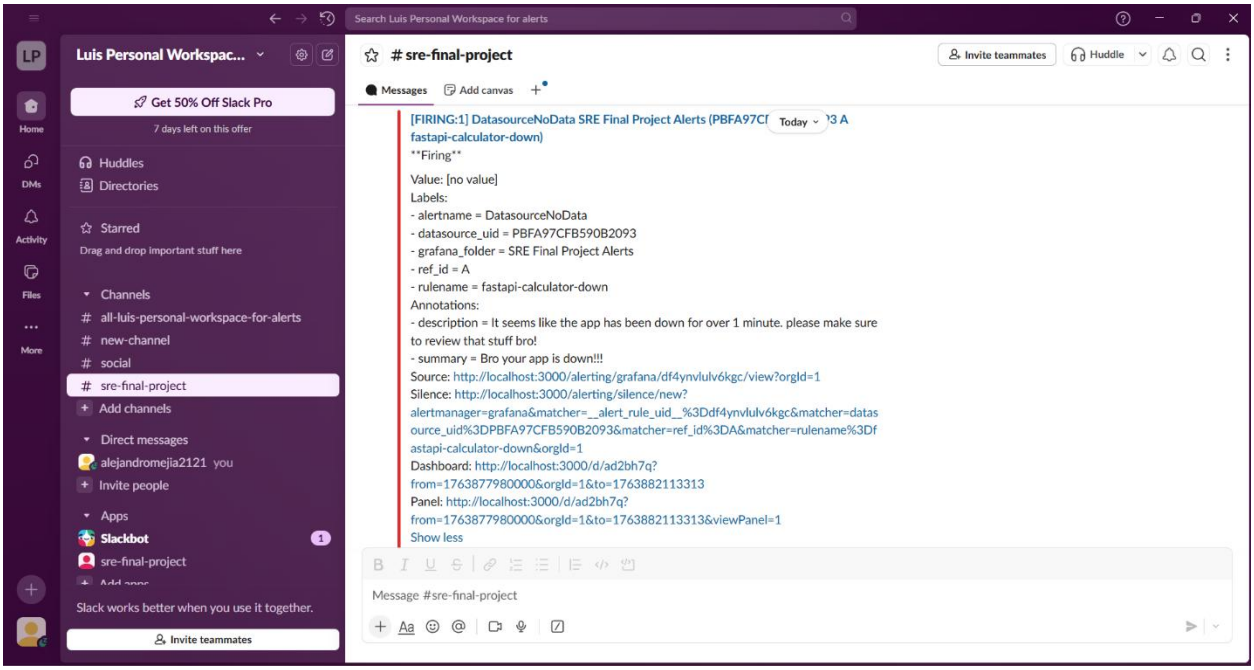
Dropped the app once again with:

kubectl scale deployment fastapi-calculator -n monitoring --replicas=0

Alert status before 1 min:



Status in Grafana did not change, but the alert was fired and received in slack:



After adjusting the rule in Grafana, decided to kill the Pods, instead of just the replicas:

Used: kubectl delete pod -n monitoring -l app=fastapi-calculator

```
Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl scale deployment fastapi-calculator -n monitoring --replicas=0
deployment.apps/fastapi-calculator scaled
Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl scale deployment fastapi-calculator -n monitoring --replicas=4
deployment.apps/fastapi-calculator scaled
Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl scale deployment fastapi-calculator -n monitoring --replicas=0
deployment.apps/fastapi-calculator scaled
Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl scale deployment fastapi-calculator -n monitoring --replicas=4
deployment.apps/fastapi-calculator scaled
Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl scale deployment fastapi-calculator -n monitoring --replicas=0
deployment.apps/fastapi-calculator scaled
Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl scale deployment fastapi-calculator -n monitoring --replicas=4
deployment.apps/fastapi-calculator scaled
Joro_21@L-Mejia21:~/projects/sre-final-project$ kubectl delete pod -n monitoring -l app=fastapi-calculator
pod "fastapi-calculator-747b9bb567-2krv4" deleted from monitoring namespace
pod "fastapi-calculator-747b9bb567-6srdx" deleted from monitoring namespace
pod "fastapi-calculator-747b9bb567-g25kd" deleted from monitoring namespace
pod "fastapi-calculator-747b9bb567-x2fg5" deleted from monitoring namespace
Joro_21@L-Mejia21:~/projects/sre-final-project$
```

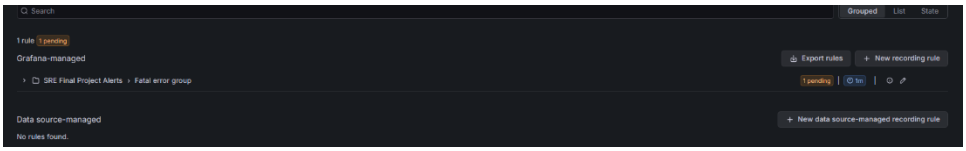
Planning to bring back with:

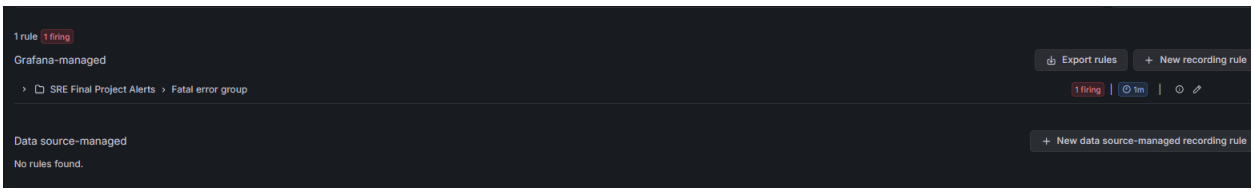
kubectl rollout restart deployment fastapi-calculator -n monitoring

But then I learnt what Kubernetes is for, because it brought it back with no issue whatsoever:

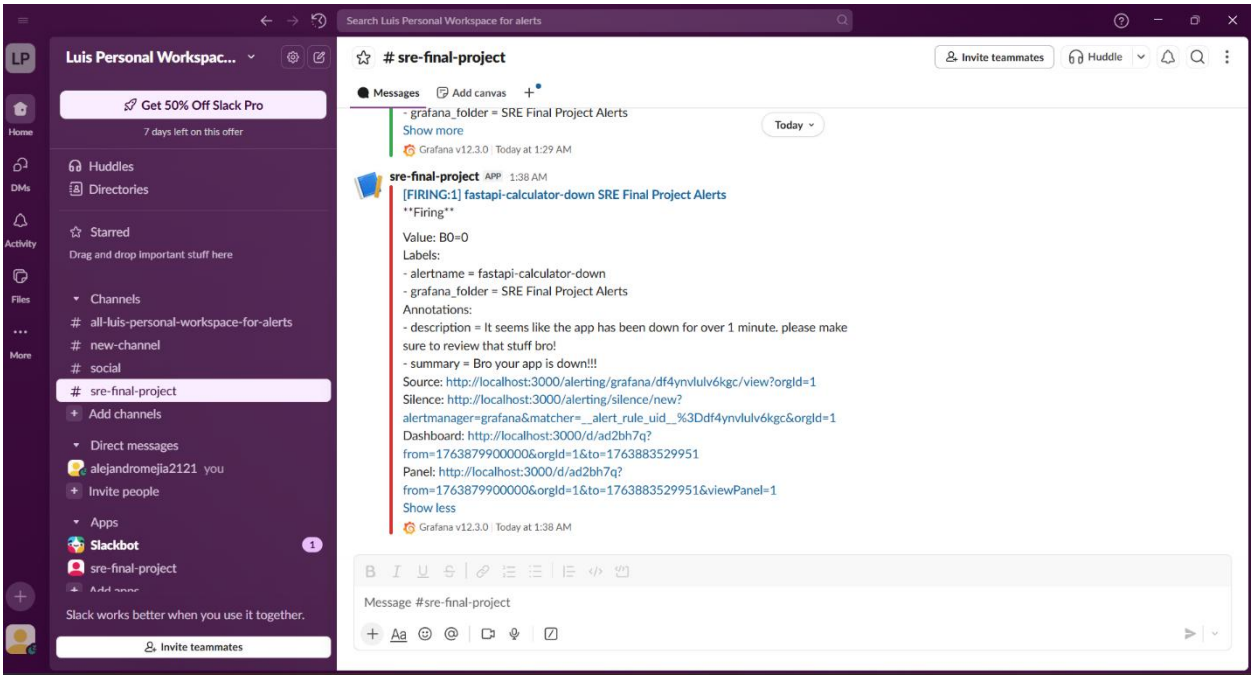
Killed the replicas again:

Grafana did respond:



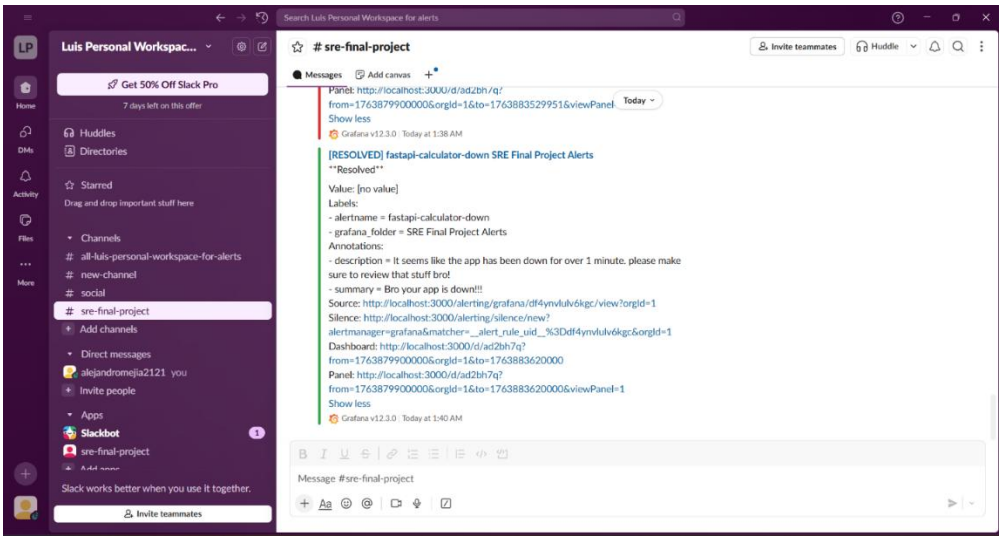


Notification sent:

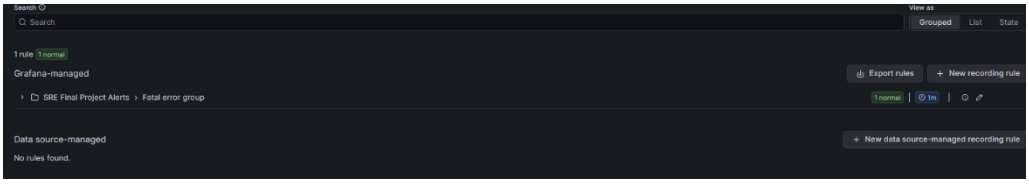


Brought it back up:

Resolved notification was received:



Alert status back to normal:



Some last touches made:

Wanted the calculator to at least have a nice entry format (with numbers and all), so I used my little knowledge in HTML to create a better view for the app. This presented a couple unexpected issues at its moment.

```
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ cd ~/projects/sre-final-project/app/
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ ls
Dockerfile  main.py  requirements.txt
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ mkdir templates
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ cd templates/
Joro_21@L-Mejia21:~/projects/sre-final-project/app/templates$ nano index.html
Joro_21@L-Mejia21:~/projects/sre-final-project/app/templates$ cd ~/projects/sre-final-project/app/
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ nano main.py
Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

```
Joro_21@L-Mejia21:~/projects/sre-final-project/app/templates$ cat index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>FastAPI Calculator</title>
    <style>
        body {
            font-family: Arial;
            background: #f5f5f5;
            display: flex;
            justify-content: center;
            margin-top: 40px;
        }
        .calc {
            background: white;
            padding: 20px;
            border-radius: 10px;
            width: 280px;
            box-shadow: 0 4px 12px rgba(0,0,0,0.15);
        }
        input, select {
            width: 100%;
            padding: 10px;
            margin: 10px 0;
        }
        button {
            width: 100%;
            padding: 12px;
            background: #0078ff;
            border: none;
            color: white;
            border-radius: 8px;
            cursor: pointer;
            font-size: 16px;
        }
        button:hover {
            background: #005fcc;
        }
    </style>
```

```
    </style>
</head>
<body>
<div class="calc">
    <h2>FastAPI Calculator</h2>
    <form action="/calculate" method="post">
        <input type="number" name="a" placeholder="First number" required>
        <input type="number" name="b" placeholder="Second number" required>

        <select name="operation">
            <option value="add">Add (+)</option>
            <option value="subtract">Subtract (-)</option>
            <option value="multiply">Multiply (×)</option>
            <option value="divide">Divide (÷)</option>
        </select>

        <button type="submit">Calculate</button>
    </form>
</div>
</body>
</html>
Joro_21@L-Mejia21:~/projects/sre-final-project/app/templates$
```

```
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ cat main.py
from fastapi import FastAPI, Form
from fastapi.responses import HTMLResponse
from fastapi.templating import Jinja2Templates
from fastapi import Request

app = FastAPI()
templates = Jinja2Templates(directory="templates")

@app.get("/", response_class=HTMLResponse)
def read_root(request: Request):
    return templates.TemplateResponse("index.html", {"request": request})

@app.post("/calculate", response_class=HTMLResponse)
def calculate(request: Request, a: float = Form(...), b: float = Form(...), operation: str = Form
)):
    if operation == "add":
        result = a + b
    elif operation == "subtract":
        result = a - b
    elif operation == "multiply":
        result = a * b
    elif operation == "divide":
        if b == 0:
            result = "Cannot divide by zero"
        else:
            result = a / b

    return templates.TemplateResponse("index.html", {"request": request, "result": result})
Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

Issues: Old image was still running and it didn't show the changes because I had not pushed the new version of the app into Kubernetes to run. However, before pushing the new image, some changes had to be made:

Changed the resources file:

fastapi

uvicorn

prometheus_client

jinja2

python-multipart

```
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ cat requirements.txt
fastapi
uvicorn
prometheus_client
jinja2
python-multipart
```

Rebuilt the docker image with a new name:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ docker build -t alejoro21/sre-calculator:1.1 .
[+] Building 38.7s (10/10) FINISHED                                                    docker:default
 => [internal] load build definition from Dockerfile                                              0.6s
 => => transferring dockerfile: 216B                                                              0.3s
 => [internal] load metadata for docker.io/library/python:3.11-slim                               3.8s
 => [auth] library/python:pull token for registry-1.docker.io                                     0.0s
 => [internal] load .dockerignore                                                                 0.0s
 => => transferring context: 2B                                                                   0.0s
 => [1/4] FROM docker.io/library/python:3.11-slim@sha256:193fdd0bbcb3d2ae612bd6cc3548d2f7c78d6    0.1s
 => => resolve docker.io/library/python:3.11-slim@sha256:193fdd0bbcb3d2ae612bd6cc3548d2f7c78d6    0.1s
 => [internal] load build context                                                                 0.0s
 => => transferring context: 3.90kB                                                               0.0s
 => CACHED [2/4] WORKDIR /app                                                                     0.0s
 => [3/4] COPY . .                                                                                0.1s
 => [4/4] RUN pip install --no-cache-dir -r requirements.txt                                     28.2s
 => exporting to image                                                                            4.8s
 => => exporting layers                                                                           2.8s
 => => exporting manifest sha256:ad95739c6bcd5a67adf18359d5a45efb3d77b8f4ea3ae718033202ac4211b    0.0s
 => => exporting config sha256:876a0d91b5b1e06cf7d01ddfde0cdad5b0729e24f69da0f3c598c64f1ec9e3a    0.0s
 => => exporting attestation manifest sha256:0b3771e0fc272e77a822a3b2ca61ea0761cda2029262ef76d    0.1s
 => => exporting manifest list sha256:d459e07b3247e637e317ab613df5b1dc03e815165fd6b9612dfebc24    0.0s
 => => naming to docker.io/alejoro21/sre-calculator:1.1                                           0.0s
 => => unpacking to docker.io/alejoro21/sre-calculator:1.1                                        1.7s
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ docker push alejoro21/sre-calculator:1.1
The push refers to repository [docker.io/alejoro21/sre-calculator]
b3ab5a841f66: Pushed
aaf07fbf5d43: Pushed
22b63e76fde1: Layer already exists
1771569cc129: Layer already exists
b3dd773c3296: Layer already exists
4da6efc9e61a: Layer already exists
2e2ece6b1d8f: Pushed
0e4bc2bd6656: Layer already exists
1.1: digest: sha256:d459e07b3247e637e317ab613df5b1dc03e815165fd6b9612dfebc240ecd0a53 size: 856
Joro_21@L-Mejia21:~/projects/sre-final-project/app$
```

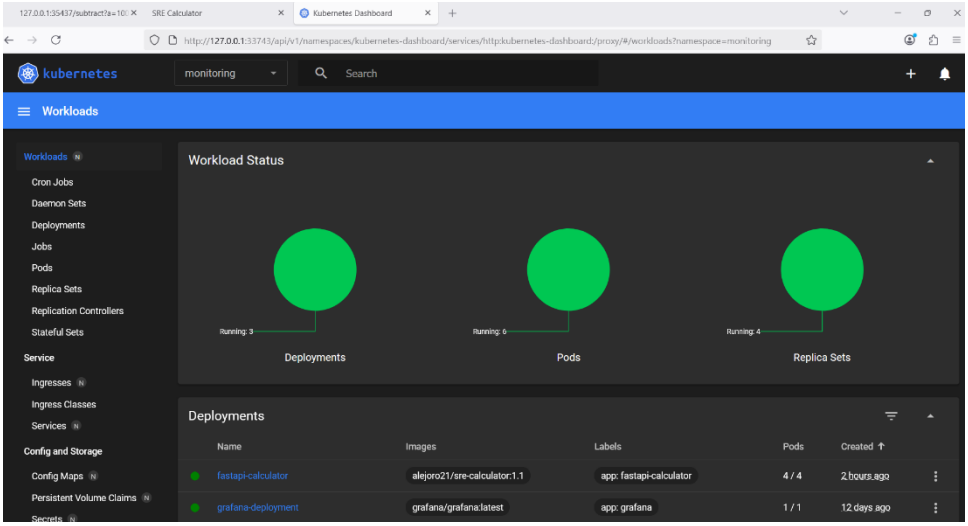
Updated Yaml file to get with the new version:

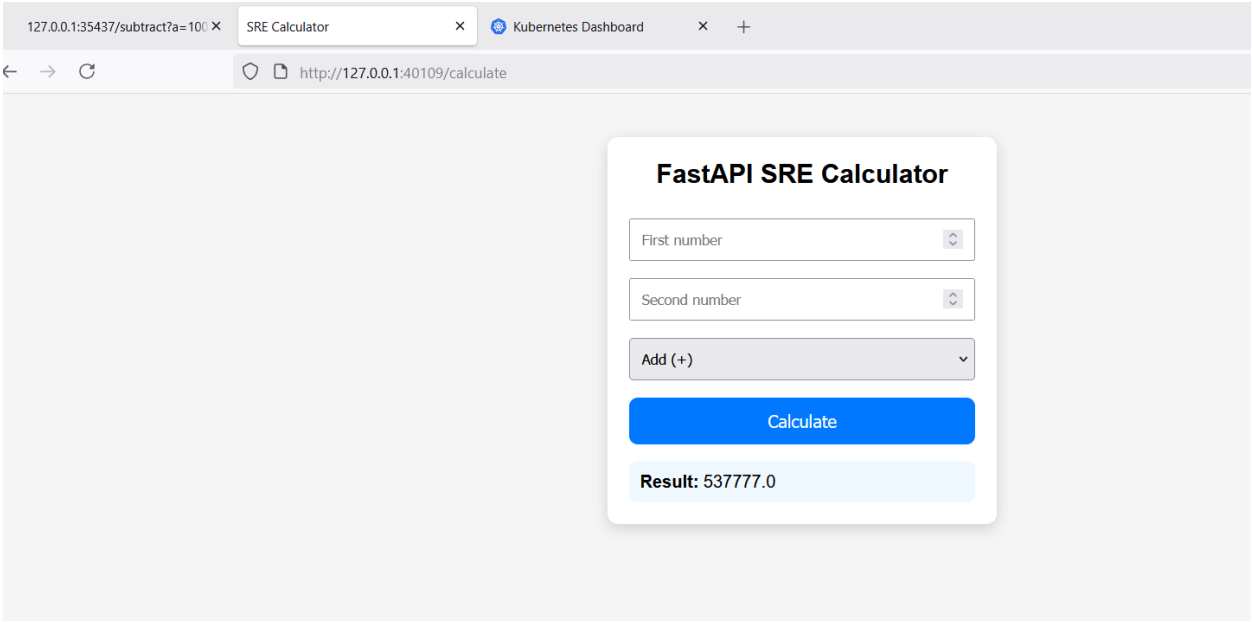
```yaml
! app.yaml 1 ×
k8s > ! app.yaml > {} spec > {} template > {} spec > [ ] containers > {} 0 > image
spec:
  template:
    metadata:
      labels:
        app: fastapi-calculator
    spec:
      containers:
        - name: fastapi
          image: alejoro21/sre-calculator:1.1
          ports:
            - containerPort: 8000
          readinessProbe:
            httpGet:
              path: /
```

Then rollout and got pods to check:

```
Joro_21@L-Mejia21:~/projects/sre-final-project/app$ cd ~/projects/sre-final-project/k8s/
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl apply -f app.yaml
namespace/monitoring unchanged
deployment.apps/fastapi-calculator configured
service/fastapi-service unchanged
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl -n monitoring rollout status deployment/fastapi-calculator
deployment "fastapi-calculator" successfully rolled out
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$ kubectl -n monitoring get pods
NAME                                  READY   STATUS    RESTARTS        AGE
fastapi-calculator-67b8964b5f-6g58b   1/1     Running   0               3m16s
fastapi-calculator-67b8964b5f-csrhn   1/1     Running   0               2m56s
fastapi-calculator-67b8964b5f-n6hcw   1/1     Running   0               2m58s
fastapi-calculator-67b8964b5f-wfmlb   1/1     Running   0               3m16s
grafana-deployment-6fd749d778-w4clk   1/1     Running   1 (127m ago)    12d
prometheus-deployment-597cbfc968-sbf2f   1/1     Running   1 (127m ago)    12d
Joro_21@L-Mejia21:~/projects/sre-final-project/k8s$
```

Verified it all run smoothly:





Looks better and it was an amazing opportunity to get hands on experience on how a real CI/CD flow could look like: (rebuild + new tag + apply).